BOUNDS FOR THE LENGTH OF FINITE PIERCE EXPANSIONS

JASON DAVIES

ABSTRACT. For integers $a > b > 0$, let $P(a,b)$ count the positive terms of the iteration $x \mapsto a \bmod x$, starting at b. We prove

$$P(a,b) \ll_\varepsilon a^{\gamma_*+\varepsilon}, \qquad \gamma_* = \frac{471}{1559} - \frac{6\sqrt{105}}{7795} = 0.2942294157\ldots.$$

Each tail of the orbit carries a finite positive measure whose Hankel determinants are positive integers. We group short segments of the orbit by small values of these determinants. Within a group, the division equations $a = q_i b_i + b_{i+1}$, kept with the fixed dividend a, confine the quotient increments to a union of boundedly many affine curves. Adjoining two further integer determinants as coordinates raises the degree of these curves while keeping the coordinates small. A curve-counting theorem whose constants do not depend on the coefficients of the curve then bounds the number of segments in each group. Raising the added coordinates to suitable powers and balancing their sizes gives the exponent $\alpha_* = (1310 - \sqrt{655})/4365$. Finally, the drop and gap rows of a segment lie in a rank-three lattice determined by its group. Coordinates in a reduced basis shrink the curvature coordinates by the square of its shortest length, and few groups have a short vector; this gives γ_*. We also prove the elementary bound $P(a,b) \ll_\varepsilon a^{3/10+\varepsilon}$, a bound for orbits with few rises in their quotient increments, and positivity and integrality of the Hankel determinants. All results, including the curve-counting input, are formalized in Lean 4 with Mathlib without proof holes or additional axioms. Shallit's conjectured $O((\log a)^2)$ bound remains open.

1. INTRODUCTION

Let $a > b > 0$ be integers. Put $b_0 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence is strictly decreasing until it reaches 0; if $b_n = 0$ we write $P(a,b) = n$. For example $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. The number $P(a,b)$ is the length of the Pierce expansion of b/a [12, 13], since

$$\frac{b}{a} = \frac{1}{q_0} - \frac{1}{q_0 q_1} + \frac{1}{q_0 q_1 q_2} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [11] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [13] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [6] combined these with a divisor argument to get $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [5] went past the exponent $1/3$, proving $P(a,b) \ll_\varepsilon a^{1/3-2/177+\varepsilon}$ with exponential sum estimates (Bourgain's exponent pair), and showed that $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Baraskar and Vukusic [1] study the related sets of iterated remainders and bound their cardinalities at each fixed iteration depth. Shallit's talk pages [14] report a bound $n^{3/10}$, described as "LLM, unpublished so far". We have not seen that argument. The preliminary $3/10$ theorem below has the same exponent up to ε, and we make no claim about how the two arguments are related.

Our strongest bound comes from the joint-invariant curve embedding of Section 9, refined by the exterior lattice coordinates of Section 10.

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Theorem 1.1. *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$, independent of a, b, such that*

$$P(a,b) \le C_\varepsilon a^{\gamma_* + \varepsilon} \qquad (0 < b < a), \qquad \gamma_* = \frac{471}{1559} - \frac{6\sqrt{105}}{7795}.$$

Here $\gamma_* = 0.2942294\ldots$ is the smaller root of $7795x^2 - 4710x + 711$. The pair of powers $(m,n) = (3,2)$ in Corollary 10.12 gives the rational exponent $153/520 = 0.2942307\ldots$, which the formalization states separately. The joint-invariant argument alone gives the following slightly weaker bound.

Theorem 1.2. *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$, independent of a, b, such that*

$$P(a,b) \le C_\varepsilon a^{(1310-\sqrt{655})/4365+\varepsilon} \qquad (0 < b < a).$$

Here $\alpha_* = (1310-\sqrt{655})/4365 = 0.2942513\ldots$. The proof produces, for each pair of positive integers (m,n), an explicit rational exponent (Corollary 9.14); the pairs $(1,1)$ and $(50,51)$ give $128/435$ and $2201/7480$, which are slightly larger than α_*. The formalization states these two instances separately, alongside Theorem 1.2. All of them are superseded by Theorem 1.1.

Positive integers along the orbit. Write $k_i = q_{i+1} - q_i$ for the quotient increments and $d_i = b_i - b_{i+1}$ for the drops. Three consecutive steps $x = b_i$, $y = b_{i+1}$, $z = b_{i+2}$, $w = b_{i+3}$ have drops d, e, f and quotient increments u, v. Their *curvature* $D = dv - eu$ satisfies

$$D\,z = df + ude + (u-1)e^2 \tag{2}$$

(Section 3). So D is a *positive* integer, even though dv and eu largely cancel, and it is small when the drops are small. Six consecutive terms carry three overlapping curvatures D, E, F and four increments u, v, w, z. In Section 5 we show that the integer

$$L = \frac{\sigma(v,w,z)\,D - \sigma(u,v,w)\,E}{v}, \qquad \sigma(u,v,w) = u^2v + (u-1)v^2 + uw,$$

satisfies the exact identity $qwL = \chi(u,v,w,z)\,E + \sigma(u,v,w)\,F$, where $q = q_i$ and χ is a polynomial with positive values (Lemma 5.1). Hence $L \ge 1$, and the quotient q is bounded by a polynomial in the small parameters u, v, w, z, E, F.

Both D and L fit into a single structure (Section 7). Each tail of the orbit carries a finite positive measure ν_i with $\int d\nu_i = b_i/a$, and the Hankel determinants $I_r(i) = a \det\bigl(\int x^{j+l}\,d\nu_i\bigr)_{j,l<r}$ are positive integers whenever the remaining orbit has at least r positive terms. The first four are b_i, d_i, D and L, in the nonterminal ranges specified in Theorem 7.2.

The joint-invariant argument. We count orbit terms in a remainder window $[A, 2A)$ and then sum over dyadic values of A; let T be the number of terms in the window. For most indices i in the window, the next few quotient increments are at most $H \asymp a/(AT)$. Then $I_4(i)$ and its neighbours are small positive integers. More precisely, the ten 3×3 minors of an explicit 3×5 integer matrix of drops and increments, which include $I_4(i)$, $I_4(i+1)$ and $I_4(i+2)$, have size at most $X \asymp A^4H^6/a^3$ (Section 8.6). Their vector is the *pattern* of i, and there are $a^{o(1)}X^5$ possible patterns.

Fix a pattern. The increments are then determined by the three *nodes* $s_j = q_{i+j} - q_i$, $j = 1, 2, 3$, which are integers in $[0, 3H]$. The nodes satisfy a quadratic equation coming from linear relations among the minors. Keeping the fixed dividend a, they also satisfy the degree-twelve equation $N_0 \operatorname{Res}(P_4, xR_4) = a\Psi^4$, where $N_0 = I_4(i)$, P_4 and R_4 are the degree-four orthogonal polynomials of two consecutive tails, and Ψ is the squared norm of P_4 (Lemma 9.1). So the nodes lie on at most 24 curves, each of degree at most 24. Counting integer points on these curves in the box $[0, 3H]^3$ directly saves at most a factor $H^{1/2}$ per curve, because a component may have degree two.

To do better, we adjoin further integer coordinates: three 2×2 minors $W=(W_0,W_1,W_2)$ of drops and increments, one of which is the curvature $D=I_3(i)$, of size at most $S\asymp A^3H^3/a^2$; and the fifth Hankel determinant $J=I_5(i)$, of size at most $Y\asymp A^5H^{10}/a^4$. On a component C of degree d these are rational functions of the nodes. At a point of C at infinity of multiplicity e, the W_j vanish to order $3e/2$ and J has a pole of order $5e/2$. A linear combination of the W_j also has one zero at an affine point, which forces one extra affine pole. Consequently the curve obtained by adjoining W_0^m,W_1^m,W_2^m,J^n to the nodes has degree at least $10n+7m$, while its coordinates are at most $\max(3H,S^m,Y^n)$. The bound of Castryck, Cluckers, Dittmann and Nguyen [3] counts the integer points of a curve of degree δ in a box of side B by $B^{1/\delta+o(1)}$, uniformly in the coefficients of the curve. Applied to the enlarged curves, it bounds the number of starts per pattern. Optimizing over the ratio n/m gives α_* (Section 9).

The box of side $\max(3H,S^m,Y^n)$ is the same for every pattern. But the drop and gap rows of a start lie in a rank-three lattice Λ_p determined by its pattern p. In a reduced basis of Λ_p the nodes have coordinates of size $H/\lambda(p)$, where $\lambda(p)$ is the length of a shortest vector, and the curvature coordinates, being 2×2 minors, have coordinates of size $S/\lambda(p)^2$. These are constant invertible linear changes of the old coordinates, so the graph degree is unchanged. Few patterns have a short vector: $\sum_p \lambda(p)^{-t}\ll a^{o(1)}X^{5-t/3}$ for $0\le t\le 1$. Optimizing again over n/m gives γ_* (Section 10).

The pattern count alone limits this method to exponents above $5/17=0.2941176\ldots$ (Section 12).

Two further bounds. The positivity of D and L already gives an elementary bound.

Theorem 1.3. *For every $\varepsilon>0$ there is a constant C_ε such that*

$$P(a,b)\le C_\varepsilon\, a^{3/10+\varepsilon}\qquad \text{for all integers } 0<b<a.$$

The proof uses only the division equations and the divisor bound $\tau(n)\ll_\delta n^\delta$. For the number T of remainders in a window $[A,2A)$ there are two bounds:

$$T(T-1)\le 4A\,\Delta, \tag{3}$$

$$T^6A^2\ll a^3\Delta, \tag{4}$$

where Δ is the largest value of the divisor function on $[1,32a^7]$. Bound (3) is the Erdős–Shallit argument. Bound (4) is new: the point $(q,x)=(\lfloor a/b_i\rfloor,b_i)$ lies on the curve $xM(q)=aN(q)+D$, with M,N explicit polynomials in (u,v) (Lemma 3.3), each such curve has $a^{o(1)}$ admissible points (Lemma 4.1), and there are $\ll (B/T)^2\cdot a/T^3$ triples (u,v,D), where $B=a/A$. Multiplying (4) by the square of (3) eliminates A and gives $T^{10}\ll a^3\Delta^3$, which is the exponent $3/10$. The six-step integer L gives (4) even without the factor Δ:

$$T^6A^2\ll a^3 \tag{5}$$

(Proposition 5.3). The only remaining loss is then the divisor factor in (3). Short-interval estimates for the Erdős–Hooley function reduce it to $(\log\log a)^{1/2}$ (see the intermediate bounds below), but no such estimate can lower the exponent $3/10$ (Section 12).

The second bound concerns the structure of long orbits. Put

$$R=1+\#\{i:0\le i\le P(a,b)-3,\ k_i<k_{i+1}\},$$

so that $R-1$ counts the rises between consecutive increments.

Theorem 1.4. *$P(a,b)\ll a^{1/4}\sqrt{R}$ uniformly for $0<b<a$. In particular, if the quotient increments never rise, then $P(a,b)\ll a^{1/4}$.*

The proof (Section 6) combines three facts. First, $D \geq 1$ says that d_i/k_i is strictly decreasing, so the drops strictly decrease along every stretch without a rise. Second, the six-step integer L gives an exact identity for six quotients in arithmetic progression, which forces their common difference to be large (Lemma 6.1). Third, a sequence of positive integers in which no value occurs too often has a large sum (Lemma 6.3). Conversely, Theorem 1.4 shows that an orbit of length about $a^{3/10}$ must contain $\gg a^{1/10}$ rises.

Intermediate bounds. The development of Theorem 1.2 passed through a sequence of weaker exponents. Each counts the patterns of Section 8, and the states belonging to each pattern, more efficiently than the last.

Exponent	Main new input	Lean declaration
$23/77 = 0.29870\ldots$	I_4 as an integer label; $O(H^2)$ states per pattern	`P_le_23_77`
$11/37 = 0.29729\ldots$	nondegenerate ternary quadratic; norm equations on slices	`P_le_11_37`
$8/27 = 0.29629\ldots$	patterns as minor vectors; $X^{5+o(1)}$ of them	`P_le_8_27`
$45/152 = 0.29605\ldots$	lattices of pivot gaps; few minor vectors with short vectors	`P_le_45_152`

All four are superseded by Theorem 1.2 and are not proved here. Their proofs are formalized in the accompanying repository. Theorem 1.2 returns to the minor-vector count of the third row and improves the count within each pattern instead, and Theorem 1.1 reuses the lattices and the short-vector count of the fourth row (Section 10). The repository also proves $P(a,b) \ll a^{3/10}\sqrt{\log\log 3a}$, with the short-sum theorem of Nair and Tenenbaum [9] and the mean-value bound of de la Bretèche and Tenenbaum [2] for the Erdős–Hooley function taken as explicit hypotheses. This bound too is superseded by Theorem 1.2.

Reading guide. Section 3 proves the curvature identity, and Section 4 the elementary bound, Theorem 1.3. Section 5 proves the six-step identity. Section 6 proves the rise bound, which is independent of the rest. Section 7 constructs the moment model and proves Theorem 7.2, and Section 8 develops the patterns and minor vectors. Section 9 proves Theorem 1.2, with routine estimates for conics and graph degrees deferred to Appendices A and B, and Section 10 refines it to Theorem 1.1. Section 11 describes the formal verification and Section 12 the limits of the method.

Formal verification. Every step of the proofs of Theorems 1.1, 1.2, 1.3, 1.4 and 7.2 has been formalized in Lean 4 [7] using Mathlib [8], without additional hypotheses. Section 11 describes the formalization. The constants in this paper are the constants in the Lean proof. The accompanying repository contains the proof source, independent submission statements and build instructions.

What is not proved. These results are far from (1), and the method cannot reach it; see Section 12.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_0 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a,b)$, so that $a > b_0 > b_1 > \cdots > b_{P-1} > 0 = b_P$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \leq b_{i+1} < b_i \qquad (0 \leq i < P).$$

An index i of the orbit is also called a *state*; quantities "at state i" are computed from $b_i, b_{i+1}, \ldots$ and $q_i, q_{i+1}, \ldots$, and are defined whenever the orbit terms they use exist. Let $\tau(n)$

be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1\le n\le 32a^7} \tau(n).$$

We write $F \ll_\eta G$ when $F \le C_\eta G$, with the constant uniform in the dividend and orbit. A factor denoted $a^{o(1)}$ is bounded by $C_\eta a^\eta$ for every fixed $\eta > 0$; such factors include the divisor losses below. We write $F \asymp G$ when both $F \ll G$ and $G \ll F$ hold. Many statements below carry explicit numerical constants; these are the constants of the Lean proof. Where a constant is large we state the bound with $\ll$ and record the constant in the proof.

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \ge q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i+1)b_{i+1} \le q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1}\rfloor \ge q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \ge 1$ there is C_k with $\tau(n)^k \le C_k\, n$ for all $n \ge 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k/p^\alpha$. Each factor with $p \ge 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \le 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \ge 1$ be an integer. A *window run* of length T is a sequence $x_0, x_1, \dots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \le a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \ge 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \ge 1$, and clearly $d, e, f \ge 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{6}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \ge 1$.

Proof. Subtracting the first two equations of (6) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \ge 1$, and $z > 0$, so $D > 0$. □

Lemma 3.3 (The fiber equation)**.** *Put $M(q) = uq^2 + (u^2+u+v)q + v$ and $N(q) = uq + u^2 + v$. Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (6), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If $x \le a$ then*

$$u\,yx \le a(d+e), \qquad v\,zy \le a(e+f), \qquad D\,z \le (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \le ad + ae$, since $qx \le a$ and $x \le a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \le a$ and $y \le a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \le df + u\,e(d+e) \le (1+u)(d+e+f)^2$. □

4. Proof of Theorem 1.3

4.1. Few blocks share a triple (u, v, D).

Lemma 4.1 (Fiber count)**.** *Let $a \ge 1$, $1 \le u, v \le a$ and $1 \le D \le 2a$. Let $M(q) = uq^2 + pq + v$ with $p = u^2 + u + v$, and let $\Lambda(q) = cq + \ell$ with $c = au$ and $\ell = a(u^2+v) + D$. Then the set*

$$Q = \{q \in \mathbb{Z}_{\ge 0} :\ xM(q) = \Lambda(q) \text{ for some integer } x \ge 1\}$$

has at most $\Delta = \max_{1 \le n \le 32a^7} \tau(n)$ elements.

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)\Lambda(q) = R \tag{7}$$

identically in q. If $q \in Q$ with witness x, substituting $\Lambda(q) = xM(q)$ into (7) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{8}$$

We bound the sizes first. From $u, v \le a$ and $D \le 2a$ we get $\ell \le 4a^3$, $p \le 3a^2$ and $c \le a^2$, so

$$|R| \le u\ell^2 + cp\ell + c^2 v \le 16a^7 + 12a^7 + a^5 \le 32a^7, \qquad c^2 \le a^4 \le 32a^7.$$

Case $R \ne 0$. By (8), $M(q)$ is a positive divisor of $|R|$. Since $u, p \ge 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \ge 0$, hence injective. Therefore $|Q| \le \tau(|R|) \le \Delta$.

Case $R = 0$. Since $M(q) > 0$, (8) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \ge 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \le \tau(c^2) \le \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and Λ might share a factor.

4.2. Counting visits to a window.

In this subsection $x_0, \dots, x_{T-1}$ is a window run in $[A, 2A)$, and $q_j = \lfloor a/x_j \rfloor$ are its quotients. We continue to write $\Delta = \max_{1 \le n \le 32a^7} \tau(n)$.

Proposition 4.2 (Drop bound; after Erdős–Shallit)**.** *If $T \ge 2$ then $T(T-1) \le 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \ge 1$ for $0 \le j \le T-2$. By telescoping, $\sum_j h_j = x_0 - x_{T-1} \le (2A-1) - A = A - 1$. Let $\mathcal{S}$ be the set of j with $h_j(T-1) \le 2A$ (short drops), and $\mathcal{L}$ the set of the other j (long drops). Each $j \in \mathcal{L}$ has $h_j(T-1) \ge 2A+1$, so

$$|\mathcal{L}|\,(2A+1) \le (T-1)\textstyle\sum_j h_j \le (T-1)(A-1).$$

If $2|\mathcal{L}| \ge T-1$ this would give $(T-1)(2A+1) \le 2(T-1)(A-1)$, which is false. Hence $2|\mathcal{L}| < T-1$, and since $|\mathcal{S}| + |\mathcal{L}| = T-1$ we get $2|\mathcal{S}| \ge T$.

For $j \in \mathcal{S}$ we have $1 \le h_j \le H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j + 1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \le 2a \le 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a+h) \le \Delta$ indices in $\mathcal{S}$ have $h_j = h$. Hence $|\mathcal{S}| \le H\Delta$, and

$$T(T-1) \le 2|\mathcal{S}|(T-1) \le 2\Delta\, H(T-1) \le 4A\Delta. \qquad \square$$

Proposition 4.3 (Curvature bound)**.** *If $T \ge 12$ then $T^6A^2 \le 995328\, a^3\Delta$.*

Proof. For $0 \le j \le T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \le 3(2A-1) - 3A = 3A - 3.$$

Let $\mathcal{S}$ be the set of j with $s_j T \le 12A$ (small blocks), and $\mathcal{L}$ the set of the other j. Then $|\mathcal{L}|(12A+1) \le T(3A-3)$. If $4|\mathcal{L}| \ge T$ this would give $T(12A+1) \le 4T(3A-3)$, which is false. So $4|\mathcal{L}| \le T-1$. Since $|\mathcal{S}| = T - 3 - |\mathcal{L}|$ and $T \ge 12$,

$$2|\mathcal{S}| \ge 2T - 6 - \tfrac{T-1}{2} \ge T.$$

Fix $j \in \mathcal{S}$. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d + e + f$. Lemma 3.4 gives $uA^2 \le uyx \le a(d+e) \le as$. Multiplying by T and using $sT \le 12A$ gives $uA^2T \le 12aA$, that is,

$$u\,AT \le 12a, \qquad \text{and likewise} \qquad v\,AT \le 12a.$$

Also $DA \le Dz \le (u+1)s^2 \le 2us^2$. Multiplying by T^2 gives $DAT^2 \le 2u(sT)^2 \le 288uA^2$, so $DT^2 \le 288uA$, and then

$$D\,T^3 \le 288\, u\, AT \le 3456\, a.$$

Thus the triple (u, v, D) of a small block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \le a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \le 2a.$$

The inequalities $U \le a$ and $D^* \le 2a$ hold because $AT \ge 12$ and $T^3 \ge 1728$.

Now fix such a triple and consider the small blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on these blocks: q_j determines $x_j = \Lambda(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ small blocks share a triple. So $|\mathcal{S}| \le U^2 D^* \Delta$, and

$$T^6A^2 = T\,(AT)^2\,T^3 \le 2\Delta\,(U\,AT)^2\,(D^*T^3) \le 2\Delta(12a)^2(3456a) = 995328\, a^3\Delta. \qquad \square$$

Corollary 4.4 (Window bound)**.** *If $T \ge 12$ then $T^{10} \le 63700992\, a^3\Delta^3$.*

Proof. Using $T \le 2(T-1)$ and the two propositions,

$$T^{10}A^2 \le 4\,(T^6A^2)\,\big(T(T-1)\big)^2 \le 4 \cdot 995328\, a^3\Delta \cdot (4A\Delta)^2 = 63700992\, a^3\Delta^3A^2.$$

Divide by A^2. $\square$

Remark 4.5. In terms of $B = a/A$, Propositions 4.2 and 4.3 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 4.4 takes the product of the two bounds, so no case split is needed.

4.3. **Conclusion of the proof.** Let $0 < b < a$. For $0 \le k \le \log_2 a$ let T_k be the number of indices $i < P$ with $2^k \le b_i < 2^{k+1}$. Every b_i with $i < P$ lies in $[1, a)$, so $P = \sum_{0 \le k \le \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

Let $\varepsilon > 0$. Choose an integer $m > 7/(5\varepsilon)$, and let C_{3m} be the constant of Lemma 2.2 with $k = 3m$. The maximum Δ is attained at some $n_0 \le 32a^7$, so $\Delta^{3m} \le C_{3m} \cdot 32a^7$. If $T_k \ge 12$, Corollary 4.4 gives

$$T_k^{10m} \le 63700992^m \, a^{3m} \, \Delta^{3m} \le K \, a^{3m+7}, \qquad K = 63700992^m \cdot C_{3m} \cdot 32,$$

and hence

$$T_k \le K^{1/(10m)} \, a^{3/10+7/(10m)} \le K^{1/(10m)} \, a^{3/10+\varepsilon/2}.$$

If $T_k \le 11$ this bound holds with 11 in place of $K^{1/(10m)}$. Finally $\lfloor \log_2 a \rfloor + 1 \le C'_\varepsilon a^{\varepsilon/2}$. Summing over k,

$$P(a,b) \le (\lfloor \log_2 a \rfloor + 1)\big(11 + K^{1/(10m)}\big) a^{3/10+\varepsilon/2} \le C_\varepsilon \, a^{3/10+\varepsilon}. \qquad \square$$

5. Six consecutive steps

Fix six consecutive terms $x_0 > x_1 > \cdots > x_5 \ge 0$ of the sequence (b_i) with $x_4 > 0$, and write their quotients as

$$\lfloor a/x_0 \rfloor = q, \quad \lfloor a/x_1 \rfloor = q + u, \quad \lfloor a/x_2 \rfloor = q + u + v,$$
$$\lfloor a/x_3 \rfloor = q + u + v + w, \quad \lfloor a/x_4 \rfloor = q + u + v + w + z.$$

(Unlike in Section 3, here w and z denote quotient increments.) By Lemma 2.1, $u, v, w, z \ge 1$. Let $f_k = x_k - x_{k+1} \ge 1$ be the drops, and let

$$D = f_0 v - f_1 u, \qquad E = f_1 w - f_2 v, \qquad F = f_2 z - f_3 w$$

be the curvatures of the blocks starting at x_0, x_1 and x_2. Write $\mathcal{E}_k$ for the division equation $x_k q_k + x_{k+1} - a = 0$, where $q_k = \lfloor a/x_k \rfloor$. Put

$$\sigma(u,v,w) = u^2 v + (u-1)v^2 + uw,$$
$$\chi(u,v,w,z) = \big(u^2 + (u-1)v\big)z + w\big(u^3 + uv(2u-1) + (u-1)v^2\big) + w^2\big((u-1)(u+v) + 1\big),$$
$$L = (vw + w^2 + z)\, D - \big(u^2 + (u-1)v\big)E - w\,(wf_0 - uf_2).$$

Lemma 5.1 (The six-step identity)**.** (1) $\sigma(v,w,z)\, D - \sigma(u,v,w)\, E = v\, L$.
(2) $q\, w\, L = \chi(u,v,w,z)\, E + \sigma(u,v,w)\, F$.

Proof. Part (1) is a polynomial identity: since $\sigma(v,w,z) \equiv -w^2$ and $\sigma(u,v,w) \equiv uw \pmod v$, and $wD + uE = v(wf_0 - uf_2)$, the difference is divisible by v, and expanding gives L. Part (2) holds modulo the division equations. Explicitly, $qwL - \chi E - \sigma(u,v,w)F$ equals

$$w(v^2 w + vw^2 + vz - w^2)\, \mathcal{E}_0 - w(u^2 w + 2uvw + uw^2 + uz + v^2 w + vw^2 - vw + vz - w^2)\, \mathcal{E}_1$$
$$+ w(u^2 v + u^2 w + uv^2 + 2uvw + uw^2 + uw + uz - v^2 - vw)\, \mathcal{E}_2 - w(u^2 v + uv^2 + uw - v^2)\, \mathcal{E}_3.$$

This is an identity of polynomials in a, q, the x_k and the increments, checked by expansion. $\square$

Corollary 5.2. *Suppose $q \ge 1$ and $u, v, w, z \le m$. Then $E, F \ge 1$, $L \ge 1$, and*

$$q \le 9m^3 E + 3m^3 F.$$

Proof. Lemma 3.2, applied to the blocks starting at x_1 and x_2, gives $E, F \ge 1$. Clearly $\chi > 0$ and $\sigma \ge 0$, so by Lemma 5.1(2) $qwL > 0$. Since L is an integer, $L \ge 1$ and $qw \le \chi E + \sigma F$. Bounding each monomial gives $\chi \le 9m^3 w$ and $\sigma(u,v,w) \le 3m^3 w$. Dividing by w gives the claim. $\square$

So the quotient q, which is about a/A, is controlled by the small parameters of a single block of six terms. No counting over many blocks is needed.

Proposition 5.3 (Six-step window bound)**.** *For a window run of length $T \geq 10$,*

$$T^6 A^2 \leq 96 \cdot 10^6 \, a^3.$$

Proof. For $0 \leq j \leq T-6$ let $s_j = x_j - x_{j+5}$. By telescoping, $\sum_{j \leq T-6} s_j \leq 5(2A-1) - 5A = 5A-5$. So some j has $s_j(T-5) \leq 5A-5$, and since $T \leq 2(T-5)$, we get $s := s_j$ with $sT \leq 10A$. Apply Corollary 5.2 to $x_j, \ldots, x_{j+5}$, all of which lie in $[A, 2A)$, with $m = \max(u, v, w, z)$.

By Lemma 3.4 at x_j, x_{j+1} and x_{j+2}, each increment satisfies (increment) $\cdot A^2 \leq as$. Multiplying by T gives $mAT \leq 10a$. By Lemma 3.4 again, $EA \leq Ex_{j+3} \leq (v+1)s^2 \leq 2ms^2$, and likewise $FA \leq 2ms^2$. Hence $qA \leq 24m^4 s^2$. Also $a = qx_j + x_{j+1} < (q+1)x_j < 2A(q+1) \leq 4Aq$. Therefore

$$qA \cdot A^4 T^6 \leq 24\,(mAT)^4 (sT)^2 \leq 24 \cdot 10^6 \, a^4 A^2,$$

so $qA^3 T^6 \leq 24 \cdot 10^6 a^4$, and $aA^2 T^6 \leq 4qA^3 T^6 \leq 96 \cdot 10^6 \, a^4$. □

Remark 5.4. In terms of $B = a/A$, Proposition 5.3 reads $T \ll a^{1/6} B^{1/3}$. This is Proposition 4.3 without the factor $\Delta^{1/6}$. Combined with Proposition 4.2 as in Corollary 4.4, it gives $T^{10} \ll a^3 \Delta^2$ and another proof of Theorem 1.3. The Lean proof of Theorem 1.3 uses Corollary 4.4.

6. Rises in the quotient increments

Throughout this section $k_i = q_{i+1} - q_i \geq 1$, $d_i = b_i - b_{i+1} \geq 1$, and a *rise* is an index $0 \leq i \leq P(a,b) - 3$ with $k_i < k_{i+1}$.

Lemma 6.1 (Six quotients in arithmetic progression)**.** *Let $q, h \geq 1$ and $x_0 > x_1 > \cdots > x_6 \geq 0$ satisfy $x_j(q + jh) + x_{j+1} = a$ for $0 \leq j \leq 5$. Then $q^4 \leq 324\, h^3 a$.*

Proof. Let $f_j = x_j - x_{j+1}$, and let $E = (f_1 - f_2)h$ and $F = (f_2 - f_3)h$ be the curvatures of the blocks starting at x_1 and x_2 (all their increments equal h). With $U = f_0 - 2f_1 + f_2$, the division equations $\mathcal{E}_j \colon x_j(q + jh) + x_{j+1} - a = 0$ give the exact identity

$$qhU = (3h-1)E + F, \qquad \text{namely} \qquad qhU - (3h-1)E - F = h(\mathcal{E}_0 - 3\mathcal{E}_1 + 3\mathcal{E}_2 - \mathcal{E}_3).$$

This is the case $u = v = w = z = h$ of Lemma 5.1, where $L = 2h^3 U$. By Lemma 3.2, $E, F \geq 1$, so the integer U is at least 1.

Now bound E and F. Subtracting consecutive division equations gives $q_j f_j \leq h x_{j+1}$, where $q_j = q + jh$. Since $x_{j+1} q_{j+1} \leq a$, this yields $q^2 f_j \leq ha$ for $1 \leq j \leq 4$. By Lemma 3.4, $Ex_3 \leq (h+1)(f_1+f_2+f_3)^2 \leq 9h^2(h+1)a^2/q^4$. Also $a < x_3(q+3h+1)$, so $Eq^4 \leq 9h^2(h+1)(q+3h+1)a$, and likewise $Fq^4 \leq 9h^2(h+1)(q+4h+1)a$. Hence

$$q^5 h \leq q^4\big((3h-1)E + F\big) \leq 27h^3(h+1)(q+4h+1)a.$$

If $q \geq h$ this gives $q^4 \leq 324 h^3 a$. If $q < h$ then $q^4 < h^4 \leq h^3 a$, since $h \leq q_1 \leq a$. □

Lemma 6.2 (Drops decrease between rises)**.** *Whenever $b_{i+2} > 0$, $d_{i+1} k_i < d_i k_{i+1}$. In particular, if $k_{i+1} \leq k_i$ then $d_{i+1} < d_i$.*

Proof. This is $D = d_i k_{i+1} - d_{i+1} k_i \geq 1$ (Lemma 3.2). If $k_{i+1} \leq k_i$ and $d_{i+1} \geq d_i$, then $d_i k_{i+1} \leq d_{i+1} k_i$, which is a contradiction. □

Lemma 6.3 (Bounded multiplicity)**.** *Let $c_j \geq 1$ be integers, $j \in J$, such that no value occurs more than M times. Then $|J|^2 \leq 2M \sum_{j \in J} c_j$.*

Proof. Induct on $|J|$. Let c_{j_0} be the largest value. The values lie in $[1, c_{j_0}]$, so $|J| \leq M c_{j_0}$. Removing j_0 and using $|J|^2 = (|J|-1)^2 + 2|J| - 1$ completes the induction. □

Proposition 6.4 (Rises in a window)**.** *Let $x_0, \dots, x_{T-1}$ be a window run in $[A, 2A)$ with $T \geq 2$, and let ρ be the number of rises among its $T-1$ increments. Then:*

(1) $(T-1)^2 \leq 2(\rho+1)(A-1)$*;*
(2) $T - 1 \leq \lfloor a/A \rfloor$*;*
(3) *if $6A \leq a$, then $T - 1 = \ell + s$ with $\ell^2 \leq 8(\rho+1)\lfloor a/A \rfloor$ and $s^3 \leq 26244\, A$.*

Proof. Split the increments into *runs*: maximal stretches containing no rise. There are at most $\rho + 1$ runs.

(1) By Lemma 6.2, the drops strictly decrease along a run, so each value of d_j occurs at most once per run, hence at most $\rho + 1$ times. Lemma 6.3 and $\sum_j d_j = x_0 - x_{T-1} \leq A - 1$ give the claim.

(2) The increments are ≥ 1, and they sum to $q_{T-1} - q_0 \leq \lfloor a/A \rfloor$.

(3) Call an increment h *low* if $26244h^3A^4 < a^3$. Five consecutive equal increments h inside the run give six quotients in arithmetic progression. Lemma 6.1 then gives $q^4 \leq 324h^3a$. On the other hand, $a < 3Aq$, which follows from $a = qx_0 + x_1$, $x_1 < x_0 < 2A$ and $6A \leq a$. Together these give $a^3 \leq 26244h^3A^4$, so h is not low. Within a run the increments are nonincreasing, so equal values are consecutive. Hence each low value occurs at most 4 times per run, and at most $4(\rho+1)$ times in all. Lemma 6.3 bounds the number ℓ of low increments by $\ell^2 \leq 8(\rho+1)\sum_j k_j \leq 8(\rho+1)\lfloor a/A \rfloor$. Let s be the number of high increments. If $s = 0$, its bound is immediate. Otherwise let h_* be their smallest value. Then $sh_* \leq \lfloor a/A \rfloor$ and $a^3 \leq 26244h_*^3A^4$, so $s^3a^3 \leq 26244A^4(a/A)^3 \leq 26244Aa^3$. □

Lemma 6.5 (Geometric sums)**.** *Let $\alpha, \beta, s, t > 0$ and $K \geq 0$. Then*

$$\sum_{k<K} \min\bigl(\alpha 2^{-sk},\ \beta 2^{tk}\bigr) \leq \Bigl(\frac{1}{1-2^{-s}} + \frac{2^t}{2^t-1}\Bigr)\alpha^{t/(s+t)}\beta^{s/(s+t)}.$$

Proof. Let $M = \alpha^{t/(s+t)}\beta^{s/(s+t)}$. Then $(\alpha 2^{-sk})^t(\beta 2^{tk})^s = M^{s+t}$, so for each k either $\beta 2^{tk} \leq M$ or $\alpha 2^{-sk} < M$. The terms $\beta 2^{tk} \leq M$ grow with ratio 2^t, so they sum to at most $M2^t/(2^t-1)$. The terms $\alpha 2^{-sk} < M$ decay with ratio 2^{-s}, so they sum to at most $M/(1-2^{-s})$. □

Proof of Theorem 1.4. The rises inside a window are rises of the orbit, so $\rho + 1 \leq R$. Combining the three parts of Proposition 6.4 gives for the window $[2^k, 2^{k+1})$ (when $6A > a$, part (2) gives $T - 1 \leq a/A < 6 \leq 30A^{1/3}$)

$$T_k \leq 1 + \sqrt{R}\,\min\bigl(\sqrt{8a}\,2^{-k/2},\ \sqrt{2}\,2^{k/2}\bigr) + \min\bigl(a\,2^{-k},\ 30 \cdot 2^{k/3}\bigr).$$

Lemma 6.5 gives $\sum_k \min(\sqrt{8a}\,2^{-k/2}, \sqrt{2}\,2^{k/2}) \ll (\sqrt{8a}\sqrt{2})^{1/2} = 2a^{1/4}$. It also gives $\sum_k \min(a2^{-k}, 30 \cdot 2^{k/3}) \ll a^{1/4}$. Since $\log_2 a + 1 \ll a^{1/4}$, summing over k gives the theorem. □

Remark 6.6. By Theorem 1.4, an orbit with $P(a,b) \asymp a^{3/10}$ must have $R \gg a^{1/10}$. Proposition 6.4(3) says more: inside the critical window $A \asymp a^{3/5}$, a run of $T \asymp a^{3/10}$ remainders must contain $\gg a^{1/5}$ rises.

7. The Hankel hierarchy

From now on we write $N = P(a,b)$, so that $b_0 > \cdots > b_{N-1} > 0 = b_N$, and $k_i = q_{i+1} - q_i$. For each remaining tail we construct a probability measure on its quotient differences. Reweighting by the reciprocal quotient will make its total mass equal to b_i/a.

Put $p_i(j) = q_{i+j} - q_i$ for $0 \le j < N - i$; the symbols p_i, w_i are local to this construction. Define the weights backwards from $w_{N-1}(0) = 1$ by

$$w_i(0) = 1 - \sum_{j=0}^{N-i-2} \frac{w_{i+1}(j)}{p_i(j+1)}, \qquad w_i(j+1) = \frac{w_{i+1}(j)}{p_i(j+1)} \quad (0 \le j < N-i-1).$$

Let μ_i be the measure with mass $w_i(j)$ at $p_i(j)$, and let

$$d\nu_i(x) = \frac{d\mu_i(x)}{q_i + x}, \qquad h_i(s) = \int x^s \, d\nu_i = \sum_{j=0}^{N-i-1} \frac{w_i(j) p_i(j)^s}{q_{i+j}}.$$

The recursion is chosen to reproduce the division equation $q_i h_i(0) + h_{i+1}(0) = 1$ for $i+1 < N$.

Lemma 7.1 (The moment model)**.** (1) *The weights are positive and sum to* 1.
(2) $h_i(0) = b_i/a$, *that is,* $b_i/a = \mathbb{E}\big[1/(q_i + X_i)\big]$ *for* $X_i \sim \mu_i$; *and* $h_{N-1}(s) = 0$ *for* $s \ge 1$.
(3) $h_i(s+1) = \sum_{t \le s} \binom{s}{t} k_i^{s-t} h_{i+1}(t)$ *for* $i + 1 < N$.
(4) $h_i(s) = (-q_i)^s b_i/a + z$ *with* $z \in \mathbb{Z}$.

Proof. (1) Since $p_i(j+1) \ge 1$, the sum defining $w_i(0)$ is at most 1. It is strictly less: if $N - i - 1 \ge 2$, the term $j = 1$ has $p_i(2) \ge 2$; if $i = N - 2$, the last gap satisfies $q_{N-1} \ge q_{N-2} + 2$, because $b_{N-1} \mid a$ gives $q_{N-1} - q_{N-2} - 1 = q_{N-2}(b_{N-2} - b_{N-1})/b_{N-1} \ge 1$. (2) Since $p_i(j+1) = q_{i+j+1} - q_i$, the definitions give $q_i h_i(0) + h_{i+1}(0) = 1$, which is the normalized division equation; and $h_{N-1}(0) = 1/q_{N-1} = b_{N-1}/a$; since $\mu_{N-1} = \delta_0$, also $h_{N-1}(s) = 0$ for $s \ge 1$. (3) Use $p_i(j+1) = k_i + p_{i+1}(j)$ and the binomial theorem. (4) For $i = N - 1$ this holds with $z = -(-q_{N-1})^s b_{N-1}/a = -(-1)^s q_{N-1}^{s-1}$ for $s \ge 1$, since $q_{N-1} b_{N-1} = a$. Then induct backwards on i using (3), $b_{i+1}/a = 1 - q_i b_i/a$ and $k_i - q_{i+1} = -q_i$. □

Theorem 7.2 (Hankel determinants)**.** *For* $1 \le r \le N - i$, *the number* $I_r(i) = a \det\big(h_i(j+l)\big)_{0 \le j,l < r}$ *is a positive integer. Moreover* $I_1(i) = b_i$, $I_2(i) = d_i$ *(if* $i + 1 < N$*),* $I_3(i) = D_i$ *(if* $i + 3 < N$*), and* $I_4(i) = L_i$ *(if* $i + 5 < N$*), where* D_i *and* L_i *are the curvature and the six-step integer of the blocks starting at* b_i.

Proof. Integrality. By Lemma 7.1(4), $h_i(s+1) + q_i h_i(s) \in \mathbb{Z}$. Multiply the Hankel matrix on the left by the unit lower triangular matrix that adds q_i times row $j-1$ to row j. This makes rows $1, \dots, r-1$ integral and leaves the determinant unchanged. Expand along row 0, whose entries are $(-q_i)^l b_i/a + (\text{integer})$; then $a \det$ is an integer.

Positivity. The Hankel matrix is the Gram matrix of $1, x, \dots, x^{r-1}$ in $L^2(\nu_i)$. Its quadratic form is $\sum_j w_i(j) q_{i+j}^{-1} \big(\sum_l c_l p_i(j)^l\big)^2$. If this vanishes, the polynomial $\sum_l c_l x^l$ vanishes at the $N - i \ge r$ distinct points $p_i(j)$, so it is zero, by the invertibility of the Vandermonde matrix. So the matrix is positive definite.

The first four. First, $I_1(i) = a h_i(0) = b_i$. For $i+1 < N$, write $\beta_j = b_{i+j}/a$. Then $h_i(0) = \beta_0$, $h_i(1) = \beta_1$ and $h_i(2) = k_i \beta_1 + \beta_2$. Hence

$$I_2(i) = a\big(\beta_0(k_i \beta_1 + \beta_2) - \beta_1^2\big) = a(\beta_0 - \beta_1) = d_i,$$

using $q_i \beta_0 + \beta_1 = q_{i+1} \beta_1 + \beta_2 = 1$. For the larger determinants, Lemma 7.1(2),(3) makes the entries $h_i(s)$, $s \le 2r - 2$, explicit polynomials in $\beta_j = b_{i+j}/a$ and $k_i, \dots, k_{i+2r-4}$; at the end of the orbit this uses $h_{N-1}(s) = 0$ for $s \ge 1$, that is, $b_N = 0$. Expanding and reducing by $q_{i+j} \beta_j + \beta_{j+1} = 1$ gives D_i and L_i in the stated ranges. The formal proof records polynomial certificates for these two reductions. □

The ranges $i + 3 < N$ and $i + 5 < N$ cannot be relaxed to all i at which D_i or L_i is defined. For $a = 926$ and $b = 229$ the orbit is $229, 10, 6, 2$, so $N = 4$, and $I_3(1) = 984$ while $D_1 = 988$.

Remark 7.3. In these index ranges, the positivity of D and L, proved earlier by explicit identities, is an instance of the positivity of Gram determinants. The integrality of L (Lemma 5.1(1)) is an instance of the integrality of I_4. A priori $I_r(i)$ involves the moments $h_i(s)$ for $s \le 2r-2$, and hence $b_i, \dots, b_{i+2r-2}$. In fact $I_3(i)$ depends only on b_i, b_{i+1}, b_{i+2} and q_i, q_{i+1}, q_{i+2}. Similarly $I_4(i)$ depends only on $b_i, \dots, b_{i+3}$ and $q_i, \dots, q_{i+4}$: the later remainders cancel.

8. Patterns

In this section we develop the patterns used in Section 9, together with the bounds on their size and number. We keep the notation of Section 7: $b_0 > b_1 > \cdots > b_{N-1} > 0$ is the positive part of the orbit, $N = P(a,b)$, and $q_i = \lfloor a/b_i \rfloor$.

8.1. **Heuristics.** The exponent $3/10$ of Theorem 1.3 comes from a single balanced configuration: $T \asymp a^{3/10}$ remainders in a window $[A, 2A)$ with $A \asymp a^{3/5}$, quotients $q \asymp a^{2/5}$ and typical increments $k \asymp a^{1/10}$. At this scale the four-term Hankel invariant $I_4(i) = L(i)$ has size $\asymp aH^6/q^4 \asymp 1$, where H bounds the increments. The idea is that a positive integer of bounded size carries information. The labels $I_4(i), I_4(i+1), I_4(i+2)$, together with four further integers of the same size, form a *pattern* (made precise in Section 8.6). States with a given pattern satisfy two linear equations and one quadratic equation in five consecutive increments (Lemma 8.1), and the quotient q_i is a root of a quartic fixed by the pattern and the increments (Lemma 8.3). So the problem becomes one of counting patterns, and the states belonging to each pattern. Counting the latter with these equations alone already improves on $3/10$ (see the intermediate bounds in Section 1). Section 9 instead keeps the fixed dividend a in the equations.

8.2. **The pattern matrix.** Fix a state i with $i + 6 < N$ and write, for $m \ge 0$,

$$k_m = q_{i+m+1} - q_{i+m}, \qquad d_m = b_{i+m} - b_{i+m+1},$$
$$s_m = k_0 + \cdots + k_{m-1}, \qquad r_m = k_m(s_m + s_{m+1}) + k_{m+1}.$$

All indices here are local to the chosen state, and the definitions apply whenever the indicated orbit entries exist; the empty sum gives $s_0 = 0$. So $s_m = q_{i+m} - q_i$, $r_0 = k_0^2 + k_1$, $r_1 = k_1(2k_0 + k_1) + k_2$, and so on. Consider the 3×4 integer matrix

$$\mathcal{M}_i = \begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ k_0 & k_1 & k_2 & k_3 \\ r_0 & r_1 & r_2 & r_3 \end{pmatrix},$$

and let $U_m(i) = (-1)^{m+1} \det(\mathcal{M}_i \text{ with column } m \text{ deleted})$ for $0 \le m \le 3$.

Lemma 8.1 (Cofactor relations)**.** (1) *$U_3(i) = L(i) = I_4(i)$ and $U_0(i) = -L(i+1) = -I_4(i+1)$, where L is the six-step integer of Lemma 5.1.*
(2) *$\sum_m U_m(i)\, d_m = \sum_m U_m(i)\, k_m = \sum_m U_m(i)\, r_m = 0$.*

Proof. (2) is the Laplace expansion of a 4×4 determinant with a repeated row. For (1), the identity $\det(\text{columns } 0,1,2) = L(i)$ holds as a polynomial identity in $d_0, d_1, d_2, k_0, \dots, k_3$, by direct expansion; and $L(i) = I_4(i)$ by Theorem 7.2. Passing from i to $i+1$ shifts the d- and k-rows by one column, and, since $s_m(i+1) = s_{m+1}(i) - k_0$, it replaces r_m by $r_{m+1} - 2k_0k_{m+1}$. This row operation does not change the determinant, so $\det(\text{columns } 1,2,3) = L(i+1)$. □

Remark 8.2. The vector $C_m = U_m/U_3$ has a meaning: $\pi_i = \sum_{m \le 3} C_m(F_{m+1} - F_m)$, with Newton polynomials $F_m(x) = \prod_{l<m}(x - s_l)$, is the monic orthogonal polynomial of degree four for the measure ν_i of Section 7. The three rows of $\mathcal{M}_i$ encode $\nu_i(\pi_i) = 0$ and $\mu_i(x\pi_i) = \mu_i(x^2\pi_i) = 0$, and $\nu_i(xf) = \mu_i(f) - q_i\nu_i(f)$ turns these into $\nu_i(x^j\pi_i) = 0$ for $j \le 3$. In particular

$U_m(i)/I_4(i)$ are tail sums of the Newton coefficients of π_i: writing $\pi_i = \sum_{l\le 4}\gamma_l F_l$, one has $U_m(i)/I_4(i) = \gamma_{m+1}+\cdots+\gamma_4$, and $\gamma_0 = -C_0 = I_4(i+1)/I_4(i)$. This interpretation is used in Section 9.

The *pattern integers* of i are the seven integers

$$\theta(i) = \big(L(i),\, L(i+1),\, L(i+2),\, U_1(i),\, U_2(i),\, U_1(i+1),\, U_2(i+1)\big),$$

and we write $\theta(i) = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$. By Lemma 8.1, $\theta(i)$ determines $U(i) = (-N_1, u_1, u_2, N_0)$ and $U(i+1) = (-N_2, v_1, v_2, N_1)$ completely, and the k-relations at i and $i+1$ read

$$N_1 k_0 = u_1 k_1 + u_2 k_2 + N_0 k_3, \qquad N_1 k_4 = N_2 k_1 - v_1 k_2 - v_2 k_3.$$

8.3. **Two identities.**

Lemma 8.3 (The normalization identity)**.** *Let π_i be the monic orthogonal polynomial of degree four for ν_i, as in the remark above, and let $H_4 = \det\big(\mu_i(x^{j+l})\big)_{j,l<4}$. Then H_4 is a polynomial in $k_0,\dots,k_4$ with 43 terms, independent of i, and*

$$I_4(i)\,\pi_i(-q_i) = a\,H_4(k_0,\dots,k_4).$$

As a polynomial in q, $I_4(i)\pi_i(-q)$ has degree four and leading coefficient $I_4(i) = L(i) \ge 1$.

Proof. Since $F_{m+1} - F_m = F_m(x)(x - s_m - 1)$, the remark gives

$$I_4(i)\pi_i(x) = \sum_{m=0}^{3} U_m(i)\,F_m(x)\,(x - s_m - 1).$$

The division equations read $b_{i+m+1} = a - (q_i + s_m)b_{i+m}$. Use them to eliminate $b_{i+1},\dots,b_{i+4}$ from both sides. The difference becomes the zero polynomial in $a, q_i, b_i, k_0,\dots,k_4$; explicit certificates are in the Lean proof. Conceptually, the identity follows from the determinant formula for π_i, since $\nu_i(x^{s+1}) + q_i\nu_i(x^s) = \mu_i(x^s)$. The leading coefficient is that of the monic π_i. □

Lemma 8.4 (Labels)**.** *Let $i + 6 < N$ and put*

$$G(k_0,\dots,k_4) = (k_0+1)\big(k_1 s_2 + s_2 + k_1 + k_2 + 1\big)\big(k_2(k_1+k_2)s_3 + e_3\big),$$

where $e_3 = s_3(k_1+k_2) + s_3k_2 + (k_1+k_2)k_2 + (k_0+2k_1+3k_2)(k_3+1) + k_3^2 + 2k_3 + k_4 + 1$. Then $L(i)\,q_i^4 \le a\,G(k_0,\dots,k_4)$. If moreover $b_i < 2A$, then $L(i)\,a^3 \le 256A^4\,G(k_0,\dots,k_4)$.

Proof. The Newton polynomials $F_0,\dots,F_3$ are monic of degrees $0,\dots,3$. So the matrix $\big(\nu_i(F_jF_l)\big)$ is $\mathcal{U}\mathcal{H}\mathcal{U}^{\mathsf{T}}$, where $\mathcal{H}$ is the Hankel matrix of ν_i and $\mathcal{U}$ is unit lower triangular. Hence it has determinant $I_4(i)/a$, and it is positive definite, as in the proof of Theorem 7.2. By Hadamard's inequality,

$$\frac{I_4(i)}{a} \le \prod_{j<4}\nu_i(F_j^2) \le \prod_{j<4}\frac{\mu_i(F_j^2)}{q_i},$$

since $\nu_i = \mu_i/(q_i + x) \le \mu_i/q_i$ on the nonnegative support. By the shift rule $\mu_i(F_j g) = \mu_{i+j}\big(g(x+s_j)\big)$ and the moments $\mu_{i+m}(1) = \mu_{i+m}(x) = 1$, $\mu_{i+m}(x^2) = k_m + 1$, $\mu_{i+m}(x^3) = k_m^2 + 2k_m + k_{m+1} + 1$, the product equals G. For the second claim, $a = q_ib_i + b_{i+1} < 2A(q_i+1) \le 4Aq_i$, so $L(i)a^4 \le 256A^4L(i)q_i^4 \le 256A^4aG$. □

8.4. **Signs.** Throughout this subsection the pattern and gaps arise from the same state, so the cofactor relations of Lemma 8.1 remain in force. Besides the polynomial σ of Section 5, put $\omega(x,y,z,w) = x^2z + (2x-1)yz + xz^2 + xw$. Both σ and ω are positive at positive integers.

Lemma 8.5 (Signs)**.** *If $N_0, N_1 \geq 1$ and $k_0, \dots, k_4 \geq 1$, then $u_1 > 0 > u_2$.*

Proof. Solving the k- and r-relations of Lemma 8.1 for (u_1, u_2) by Cramer's rule gives the exact identities

$$u_1\sigma(k_1,k_2,k_3) = N_1\omega(k_0,k_1,k_2,k_3) + N_0\sigma(k_2,k_3,k_4),$$
$$-u_2\sigma(k_1,k_2,k_3) = N_1\sigma(k_0,k_1,k_2) + N_0\omega(k_1,k_2,k_3,k_4),$$

since the Cramer determinant is $k_1r_2 - k_2r_1 = \sigma(k_1,k_2,k_3)$. □

8.5. **Good starts.**

Lemma 8.6 (Good starts)**.** *Let $b_s, \dots, b_{s+T-1}$ be a window run in $[A, 2A)$ with $T \geq 36$, and put $H = \lfloor 28a/(AT) \rfloor + 1$. Call a start i* good *if $s \leq i$, $i+9 \leq s+T$ and $q_{j+1} - q_j \leq H$ for $i \leq j \leq i+6$, that is, $k_0, \dots, k_6 \leq H$ at state i. Then at least $T/2$ starts are good, and every good start satisfies $i+8 < N$ and $b_i, b_{i+1}, b_{i+2} < 2A$.*

Proof. Let $\mathcal{B}$ be the set of $j \in [s, s+T-1)$ with $q_{j+1} - q_j > H$. Then $|\mathcal{B}|(H+1) \leq q_{s+T-1} - q_s \leq a/A$. Since $H \cdot AT > 28a$, this gives $28|\mathcal{B}| \leq T$. Each $j \in \mathcal{B}$ spoils at most seven starts, so at least $T - 9 - 7|\mathcal{B}| \geq T/2$ starts are good, as $T \geq 36$. The last assertion is immediate, since the states $i, \dots, i+8$ lie in the window. □

8.6. **The full minor vector.** The seven pattern integers are not independent: they are coordinates of a single vector of 3×3 minors. If all ten minors are at most X in absolute value, there are only $X^{5+o(1)}$ such vectors (Proposition 8.9), rather than the X^7 choices of seven independent integers. We therefore define the *pattern* of a state i to be the vector $p(i)$ of the ten 3×3 minors of the matrix $\widehat{\mathcal{M}}_i$ below; it determines the pattern integers $\theta(i)$.

Let $\widehat{\mathcal{M}}_i$ be the 3×5 matrix with rows $(d_0, \dots, d_4)$, $(k_0, \dots, k_4)$ and $(r_0, \dots, r_4)$, and let $p_{c_1c_2c_3}$ be its minor on columns $c_1 < c_2 < c_3$. Expanding the cofactors above gives

$$\theta(i) = (p_{012},\, p_{123},\, p_{234},\, p_{023},\, -p_{013},\, p_{134},\, -p_{124}).$$

Lemma 8.7 (Expansion)**.** *Let $q = q_i$ and $s_j = q_{i+j} - q$. If $i + j + 3 < N$ then*

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{1+s_j}{q^2} + \frac{s_j^2 + 2s_j + k_j + 1}{q^3} - e_j, \qquad 0 \leq e_j \leq \frac{\mu_{i+j}\big((x+s_j)^3\big)}{q^4}.$$

Proof. Apply μ_{i+j} to $\frac{1}{q+y} = \frac{1}{q} - \frac{y}{q^2} + \frac{y^2}{q^3} - \frac{y^3}{q^3(q+y)}$ with $y = s_j + x \geq 0$, using $b_{i+j}/a = \nu_{i+j}(1) = \mu_{i+j}\big(1/(q+s_j+x)\big)$, $\mu(1) = \mu(x) = 1$ and $\mu(x^2) = k+1$. □

Proposition 8.8 (All minors)**.** *Let $i+8 < N$, $b_i < 2A$ and $k_0, \dots, k_6 \leq H$ at state i. Then every minor of $\widehat{\mathcal{M}}_i$ satisfies $|p|\, a^3 \leq 3609600\, A^4H^6$.*

Proof. Subtracting the expansions of Lemma 8.7 for j and $j+1$ gives $d_j/a = k_j/q^2 - (r_j + k_j)/q^3 + e_{j+1} - e_j$. So replacing the first row by $d'_j = d_j - a(q^{-2} - q^{-3})k_j + aq^{-3}r_j = a(e_{j+1} - e_j)$ changes no minor. Now $|d'_j| \leq 235aH^3/q^4$, $|k_j| \leq H$ and $|r_j| \leq 10H^2$, so $|p| \leq 14100\, aH^6/q^4$. Finally $a \leq 4Aq$. □

8.7. **Counting minor vectors.**

Proposition 8.9 (Minor vectors)**.** *The number of vectors of minors of integer* 3×5 *matrices with all coordinates in* $[-X, X]$ *is at most* $1 + 20 \sum_{m \le X} 729\, m^4 \tau(m) \ll X^{5+o(1)}$.

Proof. For a nonzero vector p choose a pivot π with $|p_\pi| = |m|$ maximal, and let c_4, c_5 be the two columns outside π. Let u_a (v_a) be the minor with the a-th pivot column replaced by c_4 (c_5), with the sign of the column permutation. By Cramer's rule and the Plücker relations, the remaining three minors satisfy $m\, p_{w_a} = \pm(u \times v)_a$ (thirty polynomial identities, checked symbolically). So p is determined by $(\pi, \operatorname{sgn} m, |m|, u, v)$, where $|u_a|, |v_a| \le |m|$ and $|m|$ divides every coordinate of $u \times v$.

Fix $m \ge 1$ and u, and let $g = \gcd(m, u_1, u_2, u_3)$, $n = m/g$, $u = g u_0$. By Bézout there are $c \in \mathbb{Z}^3$ and $x \in \mathbb{Z}$ with $c \cdot u_0 + xn = 1$. If $m \mid u \times v$, then $n \mid u_0 \times v$, and the identity $v - (c \cdot v)u_0 = xn\, v + c \times (v \times u_0)$ shows $v \equiv (c \cdot v)u_0 \pmod n$. Hence v lies in one of n classes modulo n, each meeting $[-m, m]^3$ in at most $(3g)^3$ points: at most $27mg^2$ choices. Summing over u, grouped by $g = d \mid m$, gives $\sum_u g^2 \le \sum_{d \mid m} d^2 (3m/d)^3 \le 27 m^3 \tau(m)$, so at most $729 m^4 \tau(m)$ pairs (u, v). There are ten pivots and two signs. □

9. Joint integer invariants and the optimized bound

We now prove Theorem 1.2. Fix a window run in $[A, 2A)$ with $T \ge 36$ terms. Throughout this section constants may depend on fixed positive integers m, n (the powers of Section 9.9) and on a loss $\eta > 0$, but never on the dividend, the orbit or the pattern. The argument has six steps.

(1) For a fixed pattern, the nodes of every good start lie on at most 24 curves (Section 9.1).
(2) On these curves the quotient q_i and the curvature $I_3(i)$ are rational functions of the nodes. Components on which the reconstruction identities fail contribute $O(1)$ starts (Section 9.2).
(3) The directions at infinity of the curves are controlled by the Newton coefficients of the pattern. This makes nonexceptional components have even degree (Section 9.3), shows that exceptional patterns are few, and excludes lines (Section 9.4).
(4) The reconstructed functions, and hence two families of new integer coordinates W and J, have prescribed orders at infinity (Sections 9.5 and 9.6).
(5) Each nonexceptional conic contributes $a^{o(1)}$ starts (Section 9.7).
(6) On the remaining components, adjoining W_j^m and J^n raises the degree. The curve-counting theorem and a balancing of heights then give the exponent (Sections 9.8 and 9.9).

Notation. In this section

- $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ are the pattern integers of Section 8, with $N_0 = I_4(i)$, and bars denote their normalized versions (9);
- s_j are the nodes, k_j the gaps, P_4, R_4 the two orthogonal quartics, Ψ the quartic norm and f the pattern quadratic;
- $D_{jl} = d_j k_l - d_l k_j$ are the 2×2 minors of drops and gaps, so that $D_{01} = I_3(i)$ is the curvature of Section 3;
- Q and E_0 are the reconstructed quotient and curvature, $W = (W_0, W_1, W_2)$ and J the joint coordinates, and S, Y their height bounds;
- e is a multiplicity at infinity, d the degree of a component, δ the degree of a graph and $\delta_{\min} = 10n + 7m$.

9.1. **Good starts and the pattern curve.** Put

$$H = \left\lfloor \frac{28a}{AT} \right\rfloor + 1, \qquad X = \left\lfloor \frac{3609600A^4H^6}{a^3} \right\rfloor.$$

Let the good starts be as in Lemma 8.6; at least $T/2$ starts are good. By Proposition 8.8, the minor vector of a good start has all coordinates at most X, and $X \geq 1$ because its coordinate $p_{012} = L(i)$ is a positive integer. The pattern of a good start (Section 8.6) therefore has height at most X, and it determines $\theta(i)$. By Proposition 8.9, there are $O_\eta(a^\eta X^5)$ patterns.

The *nodes* of state i are $s_j = q_{i+j} - q_i$. They are the support points of the measure μ_i of Section 7, and the interpolation nodes of the Newton polynomials of Section 8. The three *free nodes* s_1, s_2, s_3 of a good start are integers in $[0, 3H]$. Distinct good starts with the same pattern have distinct free-node triples (Lemma 9.2 below).

For the pattern integers θ, put

$$(\bar N_1, \bar u_1, \bar u_2) = \frac{(N_1, u_1, u_2)}{N_0}, \qquad (\bar N_2, \bar v_1, \bar v_2) = \frac{(N_2, v_1, v_2)}{N_1}. \tag{9}$$

By Lemma 8.5, $\bar N_1 > 0$, $\bar u_1 > 0$ and $\bar u_2 < 0$. Treat the free nodes s_1, s_2, s_3 as coordinates and define

$$s_0 = 0, \quad s_4 = (\bar N_1 + \bar u_1)s_1 - (\bar u_1 - \bar u_2)s_2 - (\bar u_2 - 1)s_3, \quad k_j = s_{j+1} - s_j \quad (0 \leq j \leq 3),$$
$$k_4 = \bar N_2 k_1 - \bar v_1 k_2 - \bar v_2 k_3, \qquad s_5 = s_4 + k_4.$$

At a good start these are the actual nodes and gaps, by the two k-relations of Lemma 8.1; the first is equivalent to $\bar N_1 k_0 = \bar u_1 k_1 + \bar u_2 k_2 + k_3$. Let

$$\gamma = (\bar N_1,\ -\bar N_1 - \bar u_1,\ \bar u_1 - \bar u_2,\ \bar u_2 - 1,\ 1),$$

and let γ' be the same expression in $\bar N_2, \bar v_1, \bar v_2$. Put

$$P_4(x) = \sum_{j=0}^{4} \gamma_j \prod_{\ell=0}^{j-1} (x - s_\ell), \qquad R_4(x) = \sum_{j=0}^{4} \gamma'_j \prod_{\ell=1}^{j} (x - s_\ell).$$

By the remark after Lemma 8.1, at a good start P_4 is the monic orthogonal polynomial π_i of degree four for ν_i, and $R_4(x) = \pi_{i+1}(x - k_0)$ is the corresponding polynomial of the next state, written in the variable of the current state. Put $Z = xR_4$ and $f = \sum_{j=0}^{4} \gamma_j (s_j^2 + k_j)$, the *pattern quadratic*, and let

$$\Psi = \sum_{j=0}^{4} \gamma_j \big(s_j^3 + 3s_j^2 + 3s_j(1 + k_j) + (1 + k_j)^2\big) + \sum_{j=0}^{3} \gamma_j k_{j+1}$$
$$+ \bar N_2 r_1 - \bar v_1 r_2 - \bar v_2 r_3 - k_4(2s_4 + k_4),$$

where $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$ for $j = 1, 2, 3$. This is a cubic polynomial in the free nodes, the *quartic norm*.

Degrees throughout this section mean the leading coefficient of the eventual affine Hilbert function, equivalently the degree of the projective closure. "Integral" means that the defining ideal is prime over $\mathbb{Q}$; it does not assert geometric irreducibility or a rational parametrization.

Lemma 9.1 (The pattern curve)**.** (1) *At a good start i, $\Psi = \nu_i(P_4^2) > 0$, and the free nodes satisfy*

$$f = 0, \qquad N_0 \operatorname{Res}(P_4, Z) = a\Psi^4. \tag{10}$$

(2) *For every pattern with $\bar N_1, \bar u_1 > 0 > \bar u_2$, the two equations* (10) *have degrees two and at most twelve and no common factor. Their common zero set in $\mathbb{A}^3$ is the union of at most twenty-four integral affine curves over $\mathbb{Q}$, of total degree at most twenty-four.*

Proof. (1) Since $r_m = s_{m+1}^2 - s_m^2 + k_{m+1}$, summation by parts and the k-relation at state i show that $N_0 f$ is the r-relation $\sum_m U_m(i) r_m = 0$ of Lemma 8.1. Next, the shift rule $\mu_i\big(g \prod_{\ell<j}(x - s_\ell)\big) = \mu_{i+j}\big(g(x+s_j)\big)$ and the moments listed in the proof of Lemma 8.4 give $\mu_i(x^3 P_4) = \sum_j \gamma_j\big(s_j^3 + 3s_j^2 + 3s_j(1+k_j) + (1+k_j)^2 + k_{j+1}\big)$. Eliminating k_5 with the r-relation at state $i+1$ turns this into Ψ. Since $\nu_i(xg) = \mu_i(g) - q_i \nu_i(g)$ and P_4 is orthogonal to x^3, $\nu_i(P_4^2) = \nu_i(x^4 P_4) = \mu_i(x^3 P_4) = \Psi$, and this is positive as in the proof of Theorem 7.2.

For the resultant, Lemma 7.1(3) gives $\nu_i\big(xg(x)\big) = \nu_{i+1}\big(g(x+k_0)\big)$. Hence, for $j < 4$, $\nu_i(x^j Z) = \nu_{i+1}\big((x+k_0)^j \pi_{i+1}(x)\big) = 0$. So the monic quintic Z differs from the monic orthogonal polynomial $\pi^{(5)}$ of degree five for ν_i by a multiple of P_4, and $\operatorname{Res}(P_4, Z) = \operatorname{Res}(P_4, \pi^{(5)})$. Let n_j be the squared ν_i-norm of the monic orthogonal polynomial of degree j. The three-term recurrence, whose coefficients are $\beta_j = n_j/n_{j-1}$, gives

$$\operatorname{Res}(P_4, \pi^{(5)}) = \beta_1 \beta_2^2 \beta_3^3 \beta_4^4 = \frac{n_4^4}{n_0 n_1 n_2 n_3} = \frac{\Psi^4}{\det\big(\nu_i(x^{r+s})\big)_{r,s<4}} = \frac{a\Psi^4}{I_4(i)}.$$

Since $N_0 = I_4(i)$, this is the second equation.

(2) The degree bounds are immediate, since the resultant has total degree at most ten in the nodes. Note that $\gamma_1 < 0 < \gamma_2$, $\gamma_3 = \bar u_2 - 1 < -1$ and $\gamma_0, \gamma_4 > 0$. Since $s_0 = 0$ and $\sum_j \gamma_j s_j = 0$, the quadratic part of f is

$$f_2 = \gamma_1 s_1^2 + \gamma_2 s_2^2 + \gamma_3 s_3^2 + (\gamma_1 s_1 + \gamma_2 s_2 + \gamma_3 s_3)^2,$$

whose Hessian determinant is $8\gamma_1\gamma_2\gamma_3(1+\gamma_1+\gamma_2+\gamma_3) = -8\gamma_0\gamma_1\gamma_2\gamma_3 \neq 0$. A ternary quadratic form of rank three is not a product of linear forms, so f_2, and hence f, is irreducible. The degree-twelve part of the second equation is $-ac^4$ with $c = \sum_j \gamma_j s_j^3$. If f divided the second equation, the irreducible form f_2 would divide c^4, hence c. Divide c by f_2 as polynomials in s_3; the s_3^2 coefficient of f_2 is $\gamma_3(1+\gamma_3) \neq 0$. The coefficients of the remainder vanish only if $\gamma_1\gamma_2\gamma_3^2(\gamma_3^2 - 1) = 0$, which the signs above exclude. So the two equations have no common factor, their common zero set is a curve, and by Bézout's theorem its components over $\mathbb{Q}$ have total degree at most $2 \cdot 12$. □

9.2. **Reconstruction and retained components.** Two further identities distinguish the components on which the later rational functions have the required behavior. We give their definitions before stating the component bounds.

Form the 8×9 matrix of coefficients of $P_4, xP_4, x^2P_4, x^3P_4, Z, xZ, x^2Z, x^3Z$ in the basis $1, x, \dots, x^8$. Let c_j be $(-1)^j$ times the minor obtained by deleting column j, and define

$$Q = \frac{c_1 - c_2}{c_1 - c_0}, \qquad E_0 = a\,\frac{\det(c_{r+s})_{0\le r,s<3}(c_1 - c_0)^3}{(c_1^2 - c_0 c_2)^3}.$$

At a good start the matrix annihilates the moment vector $(h_i(0), \dots, h_i(8))$ and has rank eight. Indeed, the resultant in (10) is positive, so P_4 and Z are coprime. The identities $q_i h_i(0) + h_i(1) = q_i h_i(1) + h_i(2) = 1$, together with $h_i(0)h_i(2) > h_i(1)^2$, then give

$$h_i(j) = \frac{c_j(c_1 - c_0)}{c_1^2 - c_0 c_2} \quad (0 \le j \le 8), \qquad Q = q_i, \qquad E_0 = I_3(i).$$

In particular, all displayed denominators are nonzero at good starts.

To state the identities as equations on the curve, use the raw-moment polynomials

$$M_0 = 1, \qquad M_{r+1}(k_0, k_1, \dots) = \sum_{t=0}^{r} \binom{r}{t} k_0^{r-t} M_t(k_1, k_2, \dots).$$

Only $M_0, \dots, M_6$ are needed; they involve $k_0, \dots, k_4$. At good starts, $M_r = \mu_i(x^r)$ for $0 \le r \le 6$, by the weight recursion in Section 7. Set

$$H_3 = \det(M_{r+s})_{0\le r,s<3}, \qquad H_4 = \det(M_{r+s})_{0\le r,s<4}, \qquad H_{013} = \det(M_{r+s})_{\substack{r=0,1,2\\ s=0,1,3}}.$$

This H_4 is the polynomial of Lemma 8.3. With $p_3 = \bar{u}_2 - 1 - s_1 - s_2 - s_3$, the coefficient of x^3 in P_4, consider the *normalization* and *resolvent identities*

$$N_0 P_4(-Q) = aH_4, \qquad E_0 P_4(-Q) = a\big((Q - p_3)H_3 - H_{013}\big). \tag{11}$$

Both hold at good starts, where $Q = q_i$, $E_0 = I_3(i)$ and $P_4 = \pi_i$. The first is Lemma 8.3. For the second, write $v_j = h_i(j)$ and $P_4(x) = x^4 + p_3x^3 + p_2x^2 + p_1x + p_0$. Orthogonality of P_4 to $1, x, x^2$ gives $p_0v_j + p_1v_{j+1} + p_2v_{j+2} + p_3v_{j+3} + v_{j+4} = 0$ for $j < 3$, and $q_iv_j + v_{j+1} = \mu_i(x^j) = M_j$. Eliminating v_4, v_5, v_6 with the three relations, a direct expansion gives the polynomial identity

$$\det(v_{r+s})_{r,s<3}\, P_4(-q_i) = (q_i - p_3)\det(M_{r+s})_{r,s<3} - \det(M_{r+s})_{\substack{r=0,1,2\\ s=0,1,3}},$$

and $a\det(v_{r+s})_{r,s<3} = I_3(i)$. Among components containing a good start, we call a component *retained* if both equations, after clearing denominators, hold identically on it.

Lemma 9.2 (Removing components by polynomial equations)**.** *The two identities in* (11)*, after clearing denominators, have degrees at most* 166 *and* 404*, respectively. On a degree-d component, either an identity holds everywhere or it holds at* $166d$ *or* $404d$ *good starts at most, respectively. Consequently each component that is not retained contributes $O(1)$ good starts.*

Proof. Write $Q = Q_{\mathrm{n}}/Q_{\mathrm{d}}$ and $E_0 = E_{\mathrm{n}}/E_{\mathrm{d}}$ using the cofactor expressions above. Each entry of the coefficient matrix has degree at most four in the nodes, so each 8×8 cofactor has degree at most 32, and in particular at most 40, the bound used in the formal proof. Hence

$$\deg Q_{\mathrm{n}}, \deg Q_{\mathrm{d}} \le 40, \qquad \deg E_{\mathrm{n}}, \deg E_{\mathrm{d}} \le 240.$$

The latter bound follows from the 3×3 determinant and the cube in the numerator, and the cube of a quadratic cofactor expression in the denominator. The polynomial $\widehat{P} = Q_{\mathrm{d}}^4 P_4(-Q_{\mathrm{n}}/Q_{\mathrm{d}})$ has degree at most 163: in Newton form its longest product is $Q_{\mathrm{n}} \prod_{j=1}^{3}(Q_{\mathrm{n}} + s_jQ_{\mathrm{d}})$, of degree at most $40 + 3 \cdot 41$, because $s_0 = 0$. The cleared equations are

$$\begin{aligned} \Phi_N &= N_0\widehat{P} - aH_4Q_{\mathrm{d}}^4, \\ \Phi_R &= E_{\mathrm{n}}\widehat{P} - a\big((Q_{\mathrm{n}} - p_3Q_{\mathrm{d}})H_3 - H_{013}Q_{\mathrm{d}}\big)Q_{\mathrm{d}}^3E_{\mathrm{d}}. \end{aligned}$$

Here $\deg H_3 \le 3$, $\deg H_4 \le 6$ and $\deg H_{013} \le 4$. Thus $\deg \Phi_N \le 166$ and $\deg \Phi_R \le 404$.

If a polynomial of degree r does not vanish identically on an integral curve of degree d, its zero set on that curve has at most rd distinct rational points, by the polynomial-section form of Bézout's bound. Every good start satisfies both cleared equations. Moreover, the free nodes and pattern recover $q_i = Q$, and q_i strictly increases along the fixed orbit. Thus distinct good starts of this pattern give distinct free-node triples. Applying the point bound to whichever equation is not an identity proves the claim. The absolute bound $d \le 24$ makes the last estimate uniform. □

9.3. **Directions at infinity.** Call a pattern *exceptional* if two of the five Newton coefficients γ_j sum to zero. Since $\bar{N}_1 > 0$, $\bar{u}_1 > 0$ and $\bar{u}_2 < 0$, every γ_j is nonzero, and a pair sum can vanish only in one of the four cases

$$\bar{N}_1 + \bar{u}_2 = 1, \qquad \bar{N}_1 + \bar{u}_2 = 0, \qquad \bar{u}_1 = 1, \qquad \bar{N}_1 + \bar{u}_1 = 1. \tag{12}$$

Equivalently, the integer pattern coordinates satisfy $N_1 + u_2 = N_0$, $N_1 + u_2 = 0$, $u_1 = N_0$ or $N_1 + u_1 = N_0$.

Lemma 9.3 (Leading nodes and determinant orders)**.** *Let C be an integral component of positive degree of* (10)*. Let P be a closed point above infinity on the normalization of the projective closure of C, write* $\operatorname{ord} = \operatorname{ord}_P$*, and let $e > 0$ be the multiplicity of the hyperplane at infinity at P. For a local parameter ϖ at P, let z_j be the residue of $\varpi^e s_j$, $0 \le j \le 4$. Then the z_j are not all equal and satisfy*

$$\sum_{j=0}^{4} \gamma_j z_j^r = 0 \qquad (0 \le r \le 3).$$

If two of these five leading nodes coincide, the pattern is exceptional. For a nonexceptional pattern, therefore,

$$\begin{aligned} \operatorname{ord}(s_l - s_j) &= -e \quad (0 \le j < l \le 4), \\ \operatorname{ord} \operatorname{Res}(P_4, Z) &= -10e, \\ \operatorname{ord} H_3 = -3e, &\qquad \operatorname{ord} H_4 = -6e. \end{aligned}$$

Positive order denotes a zero and negative order a pole.

Proof. All nonzero rational constants have order zero. The multiplicity e can equivalently be written as

$$e = \max_{1 \le j \le 3} \{ -\operatorname{ord} s_j \} > 0.$$

Thus a polynomial of total node degree at most r has order at least $-re$. The residues z_j lie in the residue field of P; we do not require P to be rational. We have $z_0 = 0$, and at least one of z_1, z_2, z_3 is nonzero.

The leading nodes are distinct. The definitions of the nodes give the moment equations of orders zero and one. The highest-degree part of $f = 0$ gives the equation of order two. Finally, the highest-degree part of $N_0 \operatorname{Res}(P_4, Z) = a\Psi^4$ gives the equation of order three: the resultant has total degree at most ten, whereas the cubic part of Ψ is $\sum_j \gamma_j s_j^3$. Consequently

$$\sum_{j=0}^{4} \gamma_j z_j^r = 0 \qquad (0 \le r \le 3). \tag{13}$$

For the last of these, multiply the fixed-dividend equation by ϖ^{12e} and take residues; its left side vanishes and its right side is $a(\sum_j \gamma_j z_j^3)^4$.

If two nodes coincided, there would be at most four distinct values among the z_j. Grouping the sums in (13) by those values and using the corresponding Vandermonde matrix shows that the sum of the γ_j in each group is zero. Since there are five indices and at least two groups, some group has at most two elements. It cannot have one, because every γ_j is nonzero; it must therefore give a vanishing pair sum. This proves the collision assertion. For a nonexceptional pattern it is impossible, so all five nodes are distinct, and

$$\operatorname{ord}(s_l - s_j) = -e \qquad (0 \le j < l \le 4). \tag{14}$$

The determinant orders. Write $\mathcal{R} = \operatorname{Res}(P_4, Z)$. The leading homogeneous parts needed below are

$$[\mathcal{R}]_{10} = \gamma_0\gamma_1\gamma_2\gamma_3 \prod_{0\le j<l\le 4} (s_l - s_j), \tag{15}$$

$$[H_3]_3 = k_0k_1(k_0 + k_1),$$

$$[H_4]_6 = k_0k_1k_2(k_0 + k_1)(k_1 + k_2)(k_0 + k_1 + k_2). \tag{16}$$

Here $[F]_r$ denotes the homogeneous part of total node degree r. For completeness, the degree-ten formula can be checked without expanding the resultant. In the Newton basis beginning at s_1, s_2, s_3, its 4×4 Bezout matrix for P_4, R_4 has entry degrees at most $4 - \max(i, j)$, for $0 \le i, j < 4$. Only the diagonal term of its determinant can have degree ten. Its four leading diagonal entries are

$$\begin{aligned} &-\gamma_1 s_1(s_1 - s_2)(s_1 - s_3)(s_1 - s_4),\\ &-\gamma_2 s_2(s_2 - s_3)(s_2 - s_4),\\ &-\gamma_3 s_3(s_3 - s_4), \qquad -s_4. \end{aligned}$$

Their product gives the stated Vandermonde; the additional factor x in the second polynomial contributes $P_4(0) = \gamma_0$. The raw-moment recurrence gives

$$H_3 = k_0k_1(k_0 + k_1) + k_0k_2 - k_1^2,$$

and gives the displayed degree-six part of H_4, with remainder of degree at most five. These are finite polynomial identities.

Every factor in these leading products has order $-e$ by (14). The lower-degree terms have strictly larger order, so they cannot cancel the leading terms. Thus

$$\operatorname{ord}\mathcal{R} = -10e, \qquad \operatorname{ord} H_3 = -3e, \qquad \operatorname{ord} H_4 = -6e. \tag{17}$$

□

Corollary 9.4 (Even component degrees)**.** *Every positive-degree nonexceptional component of* (10) *has even degree. At each point above infinity its hyperplane multiplicity e is even, and* $\operatorname{ord}(N_0\Psi) = -5e/2$.

Proof. Taking orders in the fixed-dividend equation and using Lemma 9.3 gives $4 \operatorname{ord}\Psi = -10e$. Since orders are integers and $N_0 \neq 0$, this proves both local assertions. The degree of the curve is the sum of the multiplicities e, weighted by the residue-field degrees, so it is even. The retained identities are not needed here. □

9.4. **Exceptional patterns and line components.** Exceptional patterns may have components of odd degree, including lines, but there are fewer of them.

Lemma 9.5 (Exceptional patterns and lines)**.** *There are $O(X^4 \log(2X)) = O_\eta(a^\eta X^4)$ exceptional patterns. In the range*

$$124928A^4H^5X < a^3, \tag{18}$$

no line component contains a good start.

Proof. Each relation in (12) becomes a vanishing maximal minor after a fixed invertible integer column change. The transformed minor vector still has height $O(X)$. For example, if the original columns are $C_0, \dots, C_4$, then

$$\det(C_1, C_2 + C_0, C_3 - C_0) = p_{123} - p_{013} - p_{012} = N_1 + u_2 - N_0.$$

The other three relations follow by the same column additions and permutations.

A fixed zero minor reduces the Plücker relations to three integer vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ of height $O(X)$ with $\mathbf{x} \times \mathbf{y} = 0$ and $\mathbf{x} \cdot \mathbf{z} = \mathbf{y} \cdot \mathbf{z} = 0$. These vectors are signed rearrangements of the nine remaining minors, so they determine the minor vector. The case $\mathbf{x} = \mathbf{y} = 0$ has $O(X^3)$ choices. Otherwise write $\mathbf{x} = r\mathbf{v}$, $\mathbf{y} = s\mathbf{v}$ with $\mathbf{v}$ primitive. In a dyadic range $R \le |\mathbf{v}|_\infty < 2R$, there are $O(R^3)$ choices for $\mathbf{v}$ and $O((X/R)^2)$ for (r, s). The equation $\mathbf{v} \cdot \mathbf{z} = 0$ has $O(X^2/R)$ solutions in the box. Indeed, take a coordinate of absolute value $n = |\mathbf{v}|_\infty$. Primitivity gives n admissible pairs of the other two coordinates modulo n, and the chosen coordinate is then determined. Since $R \le n \ll X$, this gives $O(X^2/n)$ choices. Each range therefore contributes $O(X^4)$, and summing over $O(\log(2X))$ ranges proves the pattern count. The crude bound $X \le 10^{16} a^7$ absorbs its logarithm into a^η.

Now suppose that a line contained in the zero set of (10) passes through the free-node triple s of a good start, with direction v. Extend v to the five nodes linearly, as in Section 9.1, and write $s(t) = s + tv$. Both equations of (10) hold identically in t.

(1) The resultant has degree at most ten in t, so the t^{12} coefficient of the second equation, $-a\big(\sum_j \gamma_j v_j^3\big)^4$, vanishes. Then $\Psi(s(t))$ has degree at most two, so Ψ^4 has degree at most eight, and the t^{10} coefficient $[\mathcal{R}]_{10}(v)$ of the resultant, given by (15), vanishes too: two coordinates of v coincide.

(2) The direction satisfies the four moment equations of Lemma 9.3: those of orders zero and one by the definition of the nodes, order two from the t^2 coefficient of f, and order three from step (1). The grouping argument in the proof of Lemma 9.3 then shows that the pattern is exceptional, with exactly two groups of equal coordinates, of sizes two and three. After scaling, and in the order of (12), v is one of

$$(0,1,1,0,1), \quad (0,1,1,0,0), \quad (0,0,1,1,0), \quad (0,1,0,0,1).$$

(3) For each of these four directions the resultant has degree at most six in t. This follows from a row and column degree count in the Newton-basis Bezout matrix used in the proof of Lemma 9.3, in which coinciding leading nodes lower the degrees of the corresponding entries. Hence Ψ^4 has degree at most six, $\Psi(s(t))$ is affine in t, and its second difference $\Psi(s+v) + \Psi(s-v) - 2\Psi(s)$ vanishes.

(4) Expanding this second difference, and using $f = 0$ at $s \pm v$, gives respectively

$$-2\bar{N}_1 s_3, \qquad 2(\bar{N}_1 + \bar{u}_1)(k_1 - 1), \qquad 2(\bar{u}_2 - 1)(k_2 - 1), \qquad 2\big((\bar{u}_1 - \bar{u}_2) - (s_4 - s_1)\big).$$

The factors $\bar{N}_1$, $\bar{N}_1 + \bar{u}_1$ and $\bar{u}_2 - 1$ are nonzero, so respectively

$$s_3 = 0, \qquad k_1 = 1, \qquad k_2 = 1, \qquad s_4 - s_1 = \bar{u}_1 - \bar{u}_2.$$

The Lean modules `FixedDividendDirectionExtraction`, `FixedDividendLineResultant` and `FixedDividendLines` contain steps (2)–(4). The first case is impossible at a good start, where $s_3 \ge 3$. The polynomial G in Lemma 8.4 satisfies, by expansion,

$$G \le H^5(12k_1 + 464), \qquad G \le H^5(12k_2 + 464).$$

In the last obstruction, $k_1 + k_2 + k_3 = \bar{u}_1 - \bar{u}_2 = (u_1 - u_2)/N_0 \le 2X$. Thus each surviving case gives $k_1 \le 2X$ or $k_2 \le 2X$. The label bound $N_0 q_i^4 \le aG$, with $N_0 \ge 1$, now gives

$$q_i^4 \le aH^5(24X + 464).$$

But $q_i \ge a/(4A)$ and $24X + 464 \le 488X$ for $X \ge 1$. Condition (18), with $124928 = 4^4 \cdot 488$, gives the strict reverse inequality. □

9.5. **Orders of the reconstructed functions.**

Lemma 9.6 (Orders of the reconstructed functions)**.** *Let C be a retained nonexceptional component of positive degree containing a good start. At every point above infinity, with hyperplane multiplicity e, one has*

$$\operatorname{ord} Q = -\frac{3e}{2}, \qquad \operatorname{ord} E_0 = \frac{3e}{2}.$$

In particular, E_0 is a nonzero rational function vanishing at every point above infinity.

Proof. The good start guarantees that the reconstruction denominators are not identically zero on C. Thus Q and E_0 are rational functions on C, and retention makes both identities (11) identities in its function field.

The quotient order. Write $P_4(x) = x^4 + p_3x^3 + p_2x^2 + p_1x + p_0$. Its Newton expression gives $\deg p_j \le 4 - j$, and hence $\operatorname{ord} p_j \ge -(4-j)e$. The first retained identity gives

$$\operatorname{ord} P_4(-Q) = \operatorname{ord}(aH_4/N_0) = -6e.$$

If $\operatorname{ord} Q \ge -e$, every term of $P_4(-Q)$ would have order at least $-4e$, a contradiction. Put $r = \operatorname{ord} Q < -e$. For $0 \le j < 4$,

$$\operatorname{ord}\big(p_j(-Q)^j\big) - 4r \;\ge\; (4-j)(-e-r) > 0.$$

Thus Q^4 is the unique term of lowest order. It follows that $4r = -6e$, or $\operatorname{ord} Q = -3e/2$.

The reconstructed curvature order. The coefficient $p_3 = \bar{u}_2 - 1 - s_1 - s_2 - s_3$ has order at least $-e$. The raw-moment recurrence also shows that H_{013} has degree at most four: its degree-four part is $(2k_0 + k_1)k_0k_1(k_0 + k_1)$. Consequently

$$\operatorname{ord}(Q - p_3) = -3e/2, \qquad \operatorname{ord}\big((Q-p_3)H_3\big) = -9e/2 < -4e \le \operatorname{ord} H_{013}.$$

There is again a unique term of lowest order. Dividing the second retained identity by $P_4(-Q)$ therefore gives

$$\operatorname{ord} E_0 = -9e/2 - (-6e) = 3e/2.$$

The argument uses the discrete valuation on the normalization, so it also covers branches above singular points of the original curve. □

9.6. **The joint coordinates.** Put $d_j = b_{i+j} - b_{i+j+1}$ and

$$D_{01} = d_0k_1 - d_1k_0, \quad D_{02} = d_0k_2 - d_2k_0, \quad D_{12} = d_1k_2 - d_2k_1.$$

The first of these, D_{01}, is the curvature $I_3(i)$. With $s_j = q_{i+j} - q_i$ and $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$, the Hankel polynomial H_3 of Section 9.2 and its shift are

$$H_3 = k_0r_1 - k_1r_0 = \sigma(k_0, k_1, k_2),$$
$$H_3' = k_1r_2 - k_2r_1 = \sigma(k_1, k_2, k_3),$$

with σ as in Section 5. They are positive at every state at which they are defined. On a retained component, use the reconstructed function E_0 to define

$$E_{12} = \frac{H_3'E_0 - N_0k_1}{H_3}, \qquad E_{02} = \frac{k_0E_{12} + k_2E_0}{k_1}, \qquad W = (E_{12}, -E_{02}, E_0), \qquad J = N_0\Psi.$$

The determinant and cross-product identities give, identically,

$$W \cdot (k_0, k_1, k_2) = 0, \qquad W \cdot (r_0, r_1, r_2) = N_0. \tag{19}$$

At a good start these rational functions take the integer values $W = (D_{12}, -D_{02}, D_{01})$. Changing the rank-five Gram basis to $1, x, x^2, x^3, P_4$ gives

$$\det\big(\nu_i(x^{r+s})\big)_{r,s<5} = \det\big(\nu_i(x^{r+s})\big)_{r,s<4}\, \nu_i(P_4^2).$$

Consequently $J = I_5(i)$ is a positive integer at good starts. The heights of these integers are controlled by

$$S = \left\lfloor \frac{4352A^3H^3}{a^2} \right\rfloor, \qquad Y = \left\lfloor \frac{1259991040A^5H^{10}}{a^4} \right\rfloor.$$

Lemma 9.7 (Heights of the joint coordinates)**.** *At good starts, $|W_j| \le S$ and $1 \le J \le Y$.*

Proof. Write $q = q_i$, and recall $q \ge a/(4A)$ (proof of Lemma 8.4). For $j \le 3$, apply μ_{i+j} to the identity $(q+y)^{-1} = q^{-1} - yq^{-2} + y^2/(q^2(q+y))$ with $y = s_j + x$, as in Lemma 8.7. Since $\mu_{i+j}((s_j+x)^2) = s_j^2 + 2s_j + k_j + 1 \le 17H^2$, this gives

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{s_j+1}{q^2} + \epsilon_j, \qquad 0 \le \epsilon_j \le 17H^2/q^3.$$

Subtracting consecutive expansions, $d_j/a = k_j/q^2 + \delta_j$ with $|\delta_j| \le 34H^2/q^3$. In $D_{jl} = d_jk_l - d_lk_j$ the leading terms cancel, so $|D_{jl}|q^3 \le 68aH^3$, and $q \ge a/(4A)$ gives $|D_{jl}| \le S$, including the floor because the minors are integers.

For $J = I_5(i)$, argue as in the proof of Lemma 8.4 with five Newton polynomials instead of four: Hadamard's inequality and $\nu_i \le \mu_i/q$ give $Jq^5 \le a\prod_{j<5}\mu_i(F_j^2)$. By the shift rule, $\mu_i(F_j^2) = \mu_{i+j}\big(F_j(x+s_j)\big)$ is a polynomial of degree j in the gaps, and the gap bound gives $\mu_i(F_j^2) \le c_jH^j$ with $(c_0,\dots,c_4) = (1,2,10,77,799)$. Hence $Jq^5 \le 1230460\,aH^{10}$, and $q \ge a/(4A)$ gives $J \le Y$. □

Corollary 9.8 (Orders of the joint coordinates)**.** *In the setting of Lemma 9.6, the joint coordinates satisfy*

$$\operatorname{ord} W_0 = \operatorname{ord} W_2 = 3e/2, \qquad \operatorname{ord} W_1 \ge 3e/2, \qquad \operatorname{ord} J = -5e/2. \tag{20}$$

Proof. The cubic leading part of H_3' is $k_1k_2(k_1+k_2)$. By (14), it has order $-3e$, whereas the quadratic remainder $k_1k_3 - k_2^2$ has order at least $-2e$. Thus $\operatorname{ord} H_3 = \operatorname{ord} H_3' = -3e$ and $\operatorname{ord} k_1 = -e$. In the numerator defining E_{12} the two terms have different orders:

$$\operatorname{ord}(H_3'E_0) = -3e/2 < -e = \operatorname{ord}(N_0k_1).$$

Therefore $\operatorname{ord} E_{12} = 3e/2$. Each term in $k_0E_{12} + k_2E_0$ has order $e/2$, so their sum has order at least $e/2$. Dividing by k_1 gives $\operatorname{ord} E_{02} \ge 3e/2$. Cancellation here is harmless: equality for the first and third coordinates already gives $\min_j \operatorname{ord} W_j = 3e/2$. The formula for $J = N_0\Psi$ is Corollary 9.4. □

9.7. **Counting the nonexceptional conics.**

Proposition 9.9 (Conics)**.** *A nonexceptional component of degree two contains the free-node triples of $O_\eta(a^\eta)$ good starts, with an implied constant independent of the pattern and the component. Consequently all nonexceptional conics together contribute $O_\eta(a^\eta X^5)$ good starts.*

The idea is short. A component that is not retained contributes $O(1)$ good starts (Lemma 9.2). A retained conic through three free-node triples of good starts lies in a plane, and Corollary 9.4 leaves it a single rational branch at infinity, of multiplicity two. So it is a parabola, the image of a quadratic polynomial map $t \mapsto (\xi_0(t), \xi_1(t), \xi_2(t))/\Delta_c$ whose parameter t is an integer at every integer point. By Lemma 9.6, the reconstructed curvature E_0 is a rational function of t with a zero of order three at infinity, and it takes the integer value $I_3(i)$ at each good start. Its nonconstant denominator must then divide a fixed nonzero resultant at each such t, which leaves $a^{o(1)}$ possibilities. Appendix A gives the details, including the uniform bounds on the coefficients.

9.8. **The extra degree and the powered graph.** The conic count leaves retained nonexceptional components of degree at least four. Fix such a component C of degree d. The additional integer coordinates will increase its degree without increasing their heights too much. The key is one extra zero over the affine part of C.

All divisors in this argument are on the normalization $\widetilde{C}$ of the projective closure of C, over $\mathbb{Q}$. For a closed point P of $\widetilde{C}$, write $\deg P = [\kappa(P) : \mathbb{Q}]$, where $\kappa(P)$ is the residue field; thus $\deg(\sum_P a_P[P]) = \sum_P a_P \deg P$. Let $E_\infty = \sum_{P|\infty} e_P[P]$ be the pullback of the original hyperplane at infinity. Then $\sum_{P|\infty} e_P \deg P = d$. We do not require the closed points, or the branches over a rational point of C, to be rational.

Let $\mathcal{F}$ be the pole envelope of the curvature coordinates:

$$\mathcal{F} = \sum_P \max\{0, -\operatorname{ord}_P(W_0), -\operatorname{ord}_P(W_1), -\operatorname{ord}_P(W_2)\}[P],$$

with $\operatorname{ord}_P(0) = +\infty$. The orders in (20) show that $\mathcal{F}$ is supported over the affine part of C, and hence is disjoint from E_∞.

Lemma 9.10 (An additional affine zero)**.** *There is a rational linear combination ϕ of W_0, W_1, W_2 which is not the zero function and has a positive-order zero at a closed point P_0 of $\widetilde{C}$ outside E_∞.*

Proof. First, the W_j cannot all be scalar multiples, in $\mathbb{Q}(C)$, of a fixed nonzero vector $v \in \mathbb{Q}^3$. Such a relation, together with $W \cdot (k_0, k_1, k_2) = 0$, would put C in the plane $v_0k_0+v_1k_1+v_2k_2 = 0$. This is a genuine plane: the first three gaps are an invertible linear change of the three node coordinates. The pattern quadratic is irreducible, so its restriction to this plane is nonzero and has degree at most two. Every curve contained in this plane section therefore has degree at most two, contrary to $d \geq 4$.

Choose the free-node triple $z_* \in C(\mathbb{Q})$ of a good start. All the W_j are regular there, because their common reconstruction denominator does not vanish at a good start. Write $w_j = W_j(z_*)$. The identity $W \cdot (r_0, r_1, r_2) = N_0 \neq 0$ gives $w \neq 0$. Choose h with $w_h \neq 0$. At least one function $w_hW_j - w_jW_h$ is nonzero: otherwise $W = (W_h/w_h)w$, contradicting the preceding paragraph. Take such a function for ϕ. It has rational coefficients and satisfies $\phi(z_*) = 0$.

This zero survives normalization even if z_* is singular. Indeed, write $\phi = g/g_0$ near z_*, with g not identically zero on C, $g(z_*) = 0$ and $g_0(z_*) \neq 0$. The finite normalization has a closed point P_0 above z_*. In its discrete valuation ring the numerator is in the maximal ideal and the denominator is a unit, so $\operatorname{ord}_{P_0}(\phi) > 0$. Since z_* is affine, P_0 does not lie in E_∞. □

Lemma 9.11 (The curvature pole degree)**.** *The pole envelope satisfies*

$$\deg \mathcal{F} \geq \frac{3d}{2} + 1. \tag{21}$$

Proof. Use the function ϕ from Lemma 9.10. At every point over infinity, each W_j has order at least $3e_P/2$. The same is true of a rational linear combination; cancellation can only increase its zero order. Consequently

$$(\phi)_0 \;\geq\; \frac{3}{2}E_\infty + [P_0], \qquad (\phi)_\infty \;\leq\; \mathcal{F}.$$

The first divisor is integral because every e_P is even. The function ϕ is nonzero and has a zero, so its principal divisor has equal zero and pole degrees. Hence

$$\deg \mathcal{F} \;\geq\; \deg(\phi)_\infty = \deg(\phi)_0 \;\geq\; \frac{3}{2}\sum_{P|\infty} e_P \deg P + \deg P_0 \;\geq\; \frac{3d}{2} + 1.$$

The last inequality uses only $\deg P_0 \geq 1$, not that P_0 is rational. □

For positive integers m, n, let $\Gamma_{m,n} \subseteq \mathbb{A}^7$ be the closure of the graph obtained by adjoining W_0^m, W_1^m, W_2^m, J^n to the three original node coordinates. The graph is taken on the common domain of these rational functions. It is an integral curve, and projection to the first three coordinates is birational: on that domain the four remaining coordinates are uniquely determined. Thus $\Gamma_{m,n}$ and C have the same function field. Taking powers does not lose this property, because the original coordinates remain in the graph.

Lemma 9.12 (Degree of the powered graph)**.** *If $\delta = \deg \Gamma_{m,n}$, then*

$$2\delta \geq (5n+3m)d + 2m, \qquad \delta \geq 10n + 7m =: \delta_{\min}. \tag{22}$$

Moreover,

$$\delta \leq \big(247(m+n)+1\big)d.$$

For $m = n = 1$ one has the sharper upper bound $\delta \leq 247d$ and the lower bound $\delta \geq 4d+1 \geq 17$.

Proof. Lower bound. At a point P over the old infinity hyperplane, the original node coordinates have pole orders at most e_P, while J^n has pole order $5ne_P/2$ and all W_j^m vanish. Since $n \geq 1$, the pole envelope of the seven graph coordinates at P is exactly $5ne_P/2$. Over the affine part, the node coordinates and J are regular: $J = N_0\Psi$ is polynomial in the nodes. There the envelope is exactly m times the curvature envelope. The complete envelope of the graph coordinates is therefore

$$\mathcal{F}_{m,n} = \frac{5n}{2}E_\infty + m\mathcal{F}.$$

There is a single rational linear combination λ of the seven coordinates whose pole divisor is $\mathcal{F}_{m,n}$. To see this, fix a point P in the finite support of $\mathcal{F}_{m,n}$ and multiply all seven coordinates by a local uniformizer to the power $\mathcal{F}_{m,n}(P)$. Their residues in $\kappa(P)$ are not all zero. Cancellation of the leading pole is therefore a proper $\mathbb{Q}$-linear condition on the seven coefficients. The finitely many such proper subspaces do not cover $\mathbb{Q}^7$. Choosing coefficients outside their union gives the asserted simultaneous absence of cancellation, also when $\kappa(P) \neq \mathbb{Q}$.

As a linear function on $\Gamma_{m,n}$, λ has degree at most δ. For example, homogenize it on the projective closure of the graph: its pencil with the homogenizing coordinate consists of hyperplane sections of degree δ. Removing common zeros can only decrease the degree of the resulting map to $\mathbb{P}^1$. Birational projection identifies its function field with $\mathbb{Q}(C)$, so this degree is $\deg(\lambda)_\infty = \deg \mathcal{F}_{m,n}$. It follows that

$$\delta \;\geq\; \frac{5n}{2}d + m\deg\mathcal{F} \;\geq\; \frac{5n+3m}{2}d + m.$$

This is the first inequality in (22); $d \geq 4$ gives the second. The auxiliary linear combination is used only to measure degree: no bound on its coefficients enters the integer-point box.

Upper bound. This is a routine count of polynomial degrees, given in Appendix B. □

Every good start whose free-node triple lies on C gives an integer point on this graph. By Lemma 9.7, its coordinates have absolute values at most

$$\max(3H, S^m, Y^n).$$

The original node coordinates also show that distinct such node triples remain distinct after adjoining the powers.

9.9. **The window bound and the final exponents.** We use the coefficient-uniform affine-curve theorem of Castryck–Cluckers–Dittmann–Nguyen [3, Theorem 3], also stated as [4, Corollary 2.2]: an integral affine curve C over $\mathbb{Q}$ of degree δ in $\mathbb{A}^r$, with $r \geq 2$ fixed, has

$$N_{\mathrm{aff}}(C, B) \ll_r \delta^3 B^{1/\delta}(\log B + \delta), \qquad B \geq 1, \tag{23}$$

integer points in the box $[-B, B]^r$, with an implied constant independent of the coefficients of C. Its formal proof is discussed in Section 11.

Proposition 9.13 (Window bound)**.** *Fix positive integers m, n and $\eta > 0$, and put $\mathcal{B} = \max(H, S^m, Y^n)$. For a window run in $[A, 2A)$ with $T \geq 36$ satisfying* (18)*,*

$$T \ll_{m,n,\eta} a^\eta \left(X^5 + X^4 H^{1/2} + X^5 \mathcal{B}^{1/\delta_{\min}} \right). \tag{24}$$

Proof. At least $T/2$ starts are good, and for each pattern the free-node triples of its good starts are distinct (Lemma 9.2). By Lemma 9.1, these triples lie on at most twenty-four integral curves of degree at most twenty-four, and we bound the good starts on each curve, allowing overlaps. All the heights H, S, Y and X are at most a fixed power of a, so the logarithm in (23) is $O_\eta(a^\eta)$.

- *Exceptional patterns.* There are $O_\eta(a^\eta X^4)$ of them (Lemma 9.5). Under (18) no line contains a good start, so every relevant curve has degree $2 \leq \delta \leq 24$, and (23) in $\mathbb{A}^3$ with $B = 3H$ gives $O_\eta(a^\eta H^{1/2})$ good starts per pattern. This gives the term $X^4 H^{1/2}$.
- *Nonexceptional patterns: components that are not retained, and conics.* By Lemma 9.2 and Section 9.7, these contribute $O_\eta(a^\eta)$ per pattern, hence $O_\eta(a^\eta X^5)$ in all.
- *Nonexceptional patterns: the remaining components.* Lines do not occur, by Lemma 9.5, and odd degrees do not occur, by Corollary 9.4, so these components have degree $d \geq 4$. Their good starts give distinct integer points of $\Gamma_{m,n} \subseteq \mathbb{A}^7$ in a box of side $\max(3H, S^m, Y^n) \leq 3\mathcal{B}$. By Lemma 9.12, the degree δ of $\Gamma_{m,n}$ satisfies $\delta_{\min} \leq \delta \leq 24(247(m+n)+1)$, so (23) gives $O_{m,n,\eta}(a^\eta \mathcal{B}^{1/\delta_{\min}})$ good starts per component. This gives the term $X^5 \mathcal{B}^{1/\delta_{\min}}$.

The added coordinates W_j and J are functions of the free nodes on each component. They therefore introduce no further freely counted coordinates, and the pattern count is not enlarged. □

Corollary 9.14. *For fixed positive integers m, n and every $\varepsilon > 0$,*

$$P(a, b) \ll_{m,n,\varepsilon} a^{E(m,n)+\varepsilon}, \qquad E(m, n) = \max\left(\frac{15\delta_{\min} + m}{51\delta_{\min} + 3m},\ \frac{15\delta_{\min} + 6n}{51\delta_{\min} + 20n} \right),$$

where $\delta_{\min} = 10n + 7m$. In particular $E(1, 1) = 128/435$ and $E(50, 51) = 2201/7480$.

Proof. We bound the number T of terms in each dyadic window, eliminating the window scale A. If $T < 36$ there is nothing to prove. If $AT > 28a$, then Proposition 5.3 gives $T^4 \ll a$. Otherwise $HAT \leq 56a$, and Proposition 4.2 gives $T^2 \leq 8\Delta A$. Substituting $H \leq 56a/(AT)$ and then $A \geq T^2/(8\Delta)$ into the definitions of X, S and Y gives

$$H \ll \frac{\Delta a}{T^3}, \qquad X \ll \frac{\Delta^2 a^3}{T^{10}}, \qquad S \ll \frac{a}{T^3}, \qquad Y \ll \frac{\Delta^5 a^6}{T^{20}}. \tag{25}$$

Every power of Δ is $O(a^\eta)$, and we absorb such factors into $a^{O(\eta)}$.

- If (18) fails, then $124928 A^4 H^5 X \geq a^3$. With $X a^3 \leq 3609600 A^4 H^6$ this gives $A^8 H^{11} \gg a^6$, and $H \leq 56a/(AT)$ turns it into $A^3 T^{11} \ll a^5$, so $T^{17} \ll \Delta^3 a^5$.
- Otherwise Proposition 9.13 applies. By (25), its first term gives $T \ll a^{O(\eta)} a^{15}/T^{50}$, so $T^{51} \ll a^{15+O(\eta)}$. Its second term gives $T \ll a^{O(\eta)} a^{12+1/2}/T^{40+3/2}$, so $T^{85} \ll a^{25+O(\eta)}$.

- For the graph term, $T \ll a^{O(\eta)} X^5 \mathcal{B}^{1/\delta_{\min}}$ gives $T^{51\delta_{\min}} \ll a^{O(\eta)} a^{15\delta_{\min}} \mathcal{B}$. Since $H \geq 1$, $\max(H, S^m) \leq \max(H, S)^m \ll a^{O(\eta)} (a/T^3)^m$, so if $\mathcal{B} = \max(H, S^m)$ then $T^{51\delta_{\min}+3m} \ll a^{15\delta_{\min}+m+O(\eta)}$. If $\mathcal{B} = Y^n$ then $T^{51\delta_{\min}+20n} \ll a^{15\delta_{\min}+6n+O(\eta)}$.

The first three exponents all equal $5/17$, both fractions in $E(m,n)$ exceed $5/17$, and $T^4 \ll a$ is stronger still. Choosing η small in terms of ε, m and n, every window has $T \ll a^{E(m,n)+\varepsilon/2}$, and summing over the $O(\log a)$ windows proves the bound. For $m = n = 1$ the two fractions are $256/870 = 128/435$ and $261/887$, and the first is larger. For $(m,n) = (50, 51)$ the larger fraction is $2201/7480$. □

Proof of Theorem 1.2. Set $r = n/m$. The two fractions in $E(m,n)$ become

$$E_S(r) = \frac{150r + 106}{510r + 360}, \qquad E_Y(r) = \frac{156r + 105}{530r + 357}.$$

The first decreases and the second increases on $r > 0$. Their crossing satisfies $30r^2 - 10r - 21 = 0$, so $r_* = (5+\sqrt{655})/30$ and their common value is $\alpha_* = (1310-\sqrt{655})/4365$. Positive rational ratios approximate r_*; choose the fixed positive integers m, n to make $E(m,n) < \alpha_* + \varepsilon/2$, and apply Corollary 9.14 with loss $\varepsilon/2$. The powers and all constants depend only on ε, not on a, b. □

Remark 9.15. In terms of the scales, at the critical point $T \approx a^{\alpha_*}$ one has $A \approx a^{2\alpha_*}$, $H \approx S \approx a^{1-3\alpha_*} \approx a^{0.117}$, $X \approx a^{3-10\alpha_*} \approx a^{0.057}$ and $Y \approx a^{6-20\alpha_*} \approx a^{0.115}$. The pattern count $X^5 \approx a^{0.287}$ alone would give the exponent $5/17$. The graph term multiplies it by $\mathcal{B}^{1/\delta_{\min}}$, and at $n/m \approx r_*$ the two box radii S^m and Y^n balance, at $\mathcal{B} \approx a^{0.117m}$. This gives the extra factor $a^{0.117/(10r_*+7)} \approx a^{0.0068}$.

10. Exterior lattice coordinates

We now prove Theorem 1.1. Proposition 9.13 uses the same box for every pattern. But the drop and gap rows of a good start lie in a rank-three lattice determined by its pattern. In a reduced basis of that lattice the nodes have coordinates of size H/λ, where λ is the length of a shortest lattice vector, and the curvature coordinates, being 2×2 minors, have coordinates of size S/λ^2. Few patterns have a short vector, so on average this shrinks the box. The degree of the graph is unchanged, because the new coordinates are constant invertible linear changes of the old ones. We keep the notation and the conventions on constants of Section 9.

10.1. **The lattice of a pattern.** Let $p \neq 0$ be the minor vector of a 3×5 integer matrix $\mathcal{M}$, with pivot π, $m = p_\pi$, pivot columns $c_1 < c_2 < c_3$, other columns $c_4 < c_5$, and u, v as in Proposition 8.9. Let $\Lambda_p \subseteq \mathbb{Z}^3$ be the subgroup of x with $m \mid u \cdot x$ and $m \mid v \cdot x$. It contains $m\mathbb{Z}^3$. Write $|x|$ for the Euclidean norm and $|x|_\infty$ for the sup norm, and let $\lambda(p)$ be the least sup norm of a nonzero vector of Λ_p. Then $|x| \geq \lambda(p)$ for every nonzero $x \in \Lambda_p$.

Lemma 10.1 (Lifting)**.** *Let $L_c \in \mathbb{Z}^3$ be me_a if $c = c_a$ is a pivot column, u if $c = c_4$ and v if $c = c_5$. For every row x of $\mathcal{M}$ and every column c, $m\, x_c = L_c \cdot (x_{c_1}, x_{c_2}, x_{c_3})$. Moreover $\det(L_0, L_1, L_2) = m^2 p_{012}$, and $|L_c|_\infty \leq |m|$.*

Proof. The rows L_c are the rows of $\mathcal{M}^{\mathsf{T}} \operatorname{adj}(\mathcal{M}_\pi^{\mathsf{T}})$, where $\mathcal{M}_\pi$ is the pivot 3×3 block. This is Cramer's rule. Both identities are polynomial identities in the fifteen entries, checked symbolically for each of the ten pivots. The bound holds because the pivot is maximal. □

So the pivot parts $x' = (x_{c_1}, x_{c_2}, x_{c_3})$ of the rows of $\mathcal{M}$ lie in Λ_p.

Lemma 10.2 (Index)**.** *If t is a nonnegative integer and $(t+1)^3 > |m|$, then $\lambda(p) \leq t$.*

Proof. Write $g = \gcd(|m|, u)$, $n = |m|/g$ and $u = gu_0$. The proof of Proposition 8.9 gives $v = \mu u_0 + nw$ with $\mu \in \mathbb{Z}$ and $w \in \mathbb{Z}^3$. Put $\psi(x) = \big(u_0 \cdot x \bmod n,\ (\mu \lfloor u_0 \cdot x/n \rfloor + w \cdot x) \bmod g\big)$, which takes at most $ng = |m|$ values. So there are points $x \neq x'$ of $[0,t]^3$ with $\psi(x) = \psi(x')$. Then $u_0 \cdot (x - x') = n(f - f')$, where f, f' are the two floors, so $u \cdot (x - x') = \pm m(f - f')$, and $v \cdot (x - x') = n\big(\mu(f - f') + w \cdot (x - x')\big)$ with the bracket divisible by g. Hence $x - x' \in \Lambda_p$. □

Lemma 10.3 (Reduction)**.** *Let $\Lambda \subseteq \mathbb{Z}^3$ be a subgroup containing $m\mathbb{Z}^3$, $m \geq 1$. There is a basis b_1, b_2, b_3 of Λ with $\Delta = \det(b_1, b_2, b_3)$, $0 < |\Delta| \leq m^3$ and*

$$27\,|b_1|^2|b_2|^2|b_3|^2 \leq 64\,\Delta^2.$$

Hence every $x \in \Lambda$ is $\sum \xi_i b_i$ with $\xi \in \mathbb{Z}^3$ and $27|b_i|^2\xi_i^2 \leq 64|x|^2$.

Proof. Let I be the least nonzero $|\det|$ of a triple in Λ; the triple (me_1, me_2, me_3) gives $I \leq m^3$. A triple with $|\det| = I$ is a basis. Otherwise some $x \in \Lambda$ has a Cramer coordinate that is not an integer, say the first, $\det(x, b_2, b_3)/\Delta$, and subtracting the integer part of that coordinate times b_1 from x gives a triple with smaller nonzero determinant. Among such bases minimize $d_1 = |b_1|^2$, and then the Gram determinant $d_2 = d_1|b_2|^2 - \langle b_1, b_2\rangle^2$. Put $d_3 = \Delta^2$.

Choose n with $|2(\langle b_1, b_2\rangle + nd_1)| \leq d_1$. Then $(b_2 + nb_1, b_1, b_3)$ is a basis, so $d_1 \leq |b_2 + nb_1|^2$, and $d_1|b_2 + nb_1|^2 = d_2 + (\langle b_1, b_2\rangle + nd_1)^2$. Hence $3d_1^2 \leq 4d_2$.

Let $B(y,z) = d_1\langle y, z\rangle - \langle b_1, y\rangle\langle b_1, z\rangle$, so that $B(b_2, b_2) = d_2$, and Sylvester's identity gives $B(b_2,b_2)B(b_3,b_3) - B(b_2,b_3)^2 = d_1 d_3$. Choose n_2 with $|2(B(b_2,b_3) + n_2 d_2)| \leq d_2$ and put $c = b_3 + n_2 b_2$. Then (b_1, c, b_2) is a basis with first vector b_1, so $d_2 \leq d_1|c|^2 - \langle b_1, c\rangle^2 = B(c,c)$, while $d_2 B(c,c) = (B(b_2,b_3) + n_2 d_2)^2 + d_1 d_3$. Hence $3d_2^2 \leq 4d_1 d_3$.

Finally choose n_1 with $|2(\langle b_1, c\rangle + n_1 d_1)| \leq d_1$, and use the basis b_1, $b_2' = b_2 + nb_1$, $b_3' = c + n_1 b_1$. The identities above give $3d_1|b_2'|^2 \leq 4d_2$ and $4d_1 d_2|b_3'|^2 \leq d_2^2 + 4d_1 d_3 + d_1^2 d_2$. With the two inequalities this gives $9d_2|b_3'|^2 \leq 16 d_3$, and the product of the three bounds is the claim. For the coordinates, $\xi_1 \Delta = \det(x, b_2, b_3)$, and Hadamard's inequality gives $\det(x, b_2, b_3)^2 \leq |x|^2|b_2|^2|b_3|^2$. □

10.2. **Few patterns have a short vector.**

Lemma 10.4. *For $n \geq 1$ and $x \in \mathbb{Z}^3$, at most $27n^2 \gcd(n, x)$ vectors $w \in [-n, n]^3$ have $n \mid w \cdot x$. Here $\gcd(n, x) = \gcd(n, x_1, x_2, x_3)$.*

Proof. Reducing modulo n loses a factor at most 27. Let K be the set of solutions in $[0, n)^3$, $e = \gcd(n, x)$ and $c \cdot x \equiv e \pmod n$ by Bézout. The map $(w, j) \mapsto (w + jc) \bmod n$ is injective on $K \times [0, n/e)$: equal images give $n \mid (j - j')e$, so $j = j'$, and then $w = w'$. Hence $|K|\, n/e \leq n^3$. □

Proposition 10.5 (Short vectors)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-Z, Z]$, and let $\tau(m) \leq \tau_0$ for $m \leq Z$. Then for every $R \geq 1$,*

$$\#\{p \in \mathcal{S} : \lambda(p) \leq R\} \leq 393660\, R^3 Z^4 \tau_0^2.$$

Proof. If $\lambda(p) \leq R$, some nonzero $x \in [-R, R]^3$ has $m \mid u \cdot x$. For fixed x, the encoding of Proposition 8.9 gives at most $20 \sum_m N(m, x)$ such p, where $N(m, x)$ counts pairs (u, v) with $m \mid u \times v$ and $m \mid u \cdot x$. For each u there are at most $27m \gcd(m, u)^2$ vectors v. Grouping the u by $g = \gcd(m, u)$ and writing $u = gu'$ with $(m/g) \mid u' \cdot x$, Lemma 10.4 gives $\sum_u g^2 \leq 27m^2\tau(m)\gcd(m, x)$, so $N(m, x) \leq 729 m^3 \tau(m) \gcd(m, x)$. Finally $\sum_{0<|x|_\infty \leq R} \gcd(m, x) \leq 27R^3\tau(m)$. A divisor $h > R$ of m divides no nonzero x in the box, and for $h \leq R$ the multiples of h number at most $(2R/h + 1)^3$, with $h(2R/h + 1) \leq 3R$. □

Lemma 10.6 (Fractional weights)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-X, X]$, $X \geq 1$, and let $1 \leq \tau_0$ with $\tau(m) \leq \tau_0$ for $m \leq X$. Then for $0 \leq t \leq 1$,*

$$\sum_{p \in \mathcal{S}} \lambda(p)^{-t} \leq 3163861\, \tau_0^2 X^{5-t/3}.$$

Proof. Choose an integer L with $X \leq L^3 \leq 8X$. By Lemma 10.2, $1 \leq \lambda(p) \leq L$ for $p \in \mathcal{S}$. Writing $1/\lambda = 1/L + \sum_{\lambda \leq j < L} 1/(j(j+1))$ and using Proposition 10.5 with $Z = X$,

$$L \sum_{p \in \mathcal{S}} \frac{1}{\lambda(p)} \leq \#\mathcal{S} + L \sum_{j < L} \frac{393660\, j^3 X^4 \tau_0^2}{j(j+1)} \leq \#\mathcal{S} + 393660\, X^4 \tau_0^2 L^3.$$

The proof of Proposition 8.9 gives $\#\mathcal{S} \leq 14581\, \tau_0 X^5$, and $L^3 \leq 8X$, so the right side is at most $3163861\, \tau_0^2 X^5$. Since $L \geq X^{1/3}$, the sum of $1/\lambda(p)$ is at most $3163861\, \tau_0^2 X^{5-1/3}$. The function $x \mapsto x^t$ is concave, so by Jensen's inequality $\sum \lambda^{-t} \leq (\#\mathcal{S})^{1-t} \big(\sum \lambda^{-1}\big)^t$, which is the claim. □

10.3. **Exterior coordinates.**

Lemma 10.7 (Exterior coordinates)**.** *Let b_1, b_2, b_3 be a basis of $\Lambda \subseteq \mathbb{Z}^3$ as in Lemma 10.3, with $|b_i| \geq \lambda > 0$. Let $x, x' \in \Lambda$ have coordinates $\xi, \xi' \in \mathbb{Z}^3$, and put $w = x \times x'$ and $c = \xi \times \xi' \in \mathbb{Z}^3$. Then*

$$w = c_1\, b_2 \times b_3 + c_2\, b_3 \times b_1 + c_3\, b_1 \times b_2, \qquad b_i \cdot w = \Delta c_i, \qquad 27\lambda^4 c_i^2 \leq 64|w|^2.$$

Proof. The first identity follows by expanding $(\sum \xi_i b_i) \times (\sum \xi_i' b_i)$ bilinearly, and taking the dot product with b_i gives the second. By Cauchy–Schwarz and Lemma 10.3,

$$27|b_1|^2|b_2|^2|b_3|^2 c_i^2 \leq 64\Delta^2 c_i^2 = 64(b_i \cdot w)^2 \leq 64|b_i|^2|w|^2.$$

Cancelling $|b_i|^2$ and using $|b_j| \geq \lambda$ for the other two vectors gives the claim. □

The lemma replaces a bound $|\xi|, |\xi'| \ll |x|/\lambda$, which would give $|c| \ll |x||x'|/\lambda^2$, by $|c| \ll |x \times x'|/\lambda^2$. This matters because the drop row is large while the curvature minors are small.

Lemma 10.8 (All curvature minors)**.** *At a good start, all ten minors $D_{jl} = d_j k_l - d_l k_j$ with $0 \leq j < l \leq 4$ satisfy*

$$|D_{jl}| \leq S' := \left\lfloor \frac{9472 A^3 H^3}{a^2} \right\rfloor.$$

Proof. Argue as in Lemma 9.7, now for $j \leq 5$. The gaps $k_0, \dots, k_5$ are at most H, so $s_j \leq 5H$ and $\mu_{i+j}((s_j + x)^2) = s_j^2 + 2s_j + k_j + 1 \leq 37H^2$. This gives $d_j/a = k_j/q^2 + \delta_j$ with $|\delta_j| \leq 74H^2/q^3$ for $j \leq 4$, hence $|D_{jl}| q^3 \leq 148 a H^3$, and $q \geq a/(4A)$ gives the claim. □

Now fix a pattern p that occurs at a good start, and let b_1, b_2, b_3 be a basis of Λ_p as in Lemma 10.3, with $|m|$ in place of m. Write $\lambda = \lambda(p)$. At a good start i with pattern p, the pivot parts d' and k' of the drop and gap rows of $\widehat{\mathcal{M}}_i$ lie in Λ_p, with coordinates ξ and η. Put

$$c = \xi \times \eta, \qquad w = d' \times k'.$$

The entries of w are, up to sign, the minors $D_{c_a c_b}$ of the pivot columns.

Lemma 10.9 (The new coordinates)**.** *There are matrices $B \in \mathrm{GL}_3(\mathbb{Q})$ and $G \in \mathrm{GL}_3(\mathbb{Q})$, depending only on p, such that at every good start with pattern p*

$$(s_1, s_2, s_3) = B\eta, \qquad c = G\, W, \qquad |\eta|_\infty \leq \frac{3H}{\lambda}, \qquad |c|_\infty \leq \frac{8S'}{3\lambda^2}.$$

Proof. By Lemma 10.1, $m(k_0,k_1,k_2) = Fk'$, where F has rows L_0, L_1, L_2 and $\det F = m^2 p_{012} \neq 0$, since $p_{012} = L(i) \geq 1$. With $k' = (b_1\, b_2\, b_3)\eta$ and the cumulative sums $s_j = k_0 + \cdots + k_{j-1}$, this gives B. Since $|k'|^2 \leq 3H^2$ and $|b_i| \geq \lambda$, Lemma 10.3 gives $|\eta_i| \leq 3H/\lambda$.

For the curvature, the Binet–Cauchy identity $(\ell \cdot x)(\ell' \cdot x') - (\ell' \cdot x)(\ell \cdot x') = (\ell \times \ell') \cdot (x \times x')$ applied to the rows of F gives $m^2 W = \operatorname{Cof}(F)\, w$, where $\operatorname{Cof}(F)$ has rows $L_1 \times L_2$, $L_2 \times L_0$, $L_0 \times L_1$ and determinant $(\det F)^2 \neq 0$. Lemma 10.7 gives $\Delta c = (b_1\, b_2\, b_3)^{\mathsf{T}} w$. Together, $c = m^2 \Delta^{-1} (b_1\, b_2\, b_3)^{\mathsf{T}} \operatorname{Cof}(F)^{-1} W$, which defines G. Finally $|w|^2 \leq 3S'^2$ by Lemma 10.8, and Lemma 10.7 gives $3\lambda^2 |c_i| \leq 8S'$. □

10.4. **The lattice graph.** On a retained component C of degree $d \geq 4$ of a pattern p, let $\Gamma'_{m,n}$ be the image of the graph of

$$\big(s_1, s_2, s_3, (GW)_0^m, (GW)_1^m, (GW)_2^m, J^n\big)$$

under the linear map that replaces the nodes s by $\eta = B^{-1} s$.

Lemma 10.10 (Degree of the lattice graph)**.** *The degree δ' of $\Gamma'_{m,n}$ satisfies $\delta_{\min} \leq \delta' \leq \big(247(m+n)+1\big)d$.*

Proof. An invertible linear change of the node coordinates preserves the affine Hilbert function, so we may keep the nodes s. The proof of Lemma 9.12 uses the curvature coordinates only through the pole envelope $\mathcal{F}$, through the points at which some coordinate attains it, and through the degrees of their numerators. For a constant invertible matrix G and every point P,

$$\min_j \operatorname{ord}_P (GW)_j = \min_j \operatorname{ord}_P W_j :$$

each $(GW)_j$ is a constant combination of the W_l, and each W_l is a constant combination of the $(GW)_j$. So the pole envelope of GW is $\mathcal{F}$, and it is attained by some $(GW)_j$ at every point of its support. The numerators of GW are constant combinations of those of W, with the same degree bounds. The proof of Lemma 9.12 therefore applies unchanged. □

10.5. **The window bound and the optimized exponent.**

Proposition 10.11 (Lattice window bound)**.** *Fix positive integers m, n and $\eta > 0$, and put $h = \delta_{\min} = 10n + 7m$. For a window run in $[A, 2A)$ with $T \geq 36$ satisfying* (18)*,*

$$T \ll_{m,n,\eta} a^\eta \left(X^5 + X^4 H^{1/2} + X^{5-1/(3h)} H^{1/h} + X^{5-2m/(3h)} S'^{\,m/h} + X^5 Y^{n/h} \right).$$

Proof. The proof of Proposition 9.13 applies, except for the remaining components of the nonexceptional patterns. For such a component of a pattern p, the good starts give integer points (η, c^m, J^n) of $\Gamma'_{m,n}$, and distinct node triples give distinct points because η determines s. By Lemma 10.9 they lie in the box of side

$$\mathcal{B}_p = \max\Big(1, \frac{3H}{\lambda(p)}, \Big(\frac{8S'}{3\lambda(p)^2}\Big)^m, Y^n\Big).$$

By Lemma 10.10 and (23), there are $O_{m,n,\eta}(a^\eta \mathcal{B}_p^{1/h})$ of them, and

$$\mathcal{B}_p^{1/h} \leq 1 + 3H^{1/h} \lambda(p)^{-1/h} + 8S'^{\,m/h} \lambda(p)^{-2m/h} + Y^{n/h}.$$

Sum over the $O_\eta(a^\eta X^5)$ patterns. Since $h \geq 2m$, both $1/h$ and $2m/h$ lie in $[0,1]$, and Lemma 10.6, with $\tau_0 = a^{o(1)}$, gives $\sum_p \lambda(p)^{-1/h} \ll a^\eta X^{5-1/(3h)}$ and $\sum_p \lambda(p)^{-2m/h} \ll a^\eta X^{5-2m/(3h)}$. □

Corollary 10.12. *For fixed positive integers m, n, with $h = 10n + 7m$, and every $\varepsilon > 0$,*

$$P(a,b) \ll_{m,n,\varepsilon} a^{E'(m,n)+\varepsilon}, \qquad E'(m,n) = \max\left(\frac{5}{17},\ \frac{45h}{153h-1},\ \frac{45h-3m}{153h-11m},\ \frac{15h+6n}{51h+20n}\right).$$

In particular $E'(3,2) = 153/520$.

Proof. Argue as in the proof of Corollary 9.14, with $S' \ll a/T^3$ in place of S in (25). The first two terms and the windows in which (18) fails give the exponent $5/17$. For the third term, $T^{3h} \ll a^{O(\eta)} X^{15h-1} H^3$, and (25) gives $T^{153h-1} \ll a^{45h+O(\eta)}$. For the fourth, $T^{3h} \ll a^{O(\eta)} X^{15h-2m} S'^{3m}$ gives $T^{153h-11m} \ll a^{45h-3m+O(\eta)}$. For the fifth, $T^h \ll a^{O(\eta)} X^{5h} Y^n$ gives $T^{51h+20n} \ll a^{15h+6n+O(\eta)}$. For $(m,n) = (3,2)$, $h = 41$ and the last three fractions are $1845/6272$, $153/520$ and $627/2131$. □

Proof of Theorem 1.1. Set $r = n/m$. The last two fractions in $E'(m,n)$ become

$$E'_S(r) = \frac{450r + 312}{1530r + 1060}, \qquad E_Y(r) = \frac{156r + 105}{530r + 357},$$

where E_Y is the same function as in the proof of Theorem 1.2. The first decreases and the second increases on $r > 0$. Their crossing satisfies $15r^2 = 7$, so $r_* = \sqrt{105}/15$, and their common value is $\gamma_* = 471/1559 - 6\sqrt{105}/7795$. The fraction $45h/(153h-1)$ decreases in h, and for $h \geq 102$ it is at most $4590/15605 < \gamma_*$; also $5/17 < \gamma_*$. Choose a positive rational r close to r_* with $\max(E'_S(r), E_Y(r)) < \gamma_* + \varepsilon/2$, write $r = n_0/m_0$, and take $(m,n) = (6m_0, 6n_0)$, so that $h \geq 102$. Then $E'(m,n) < \gamma_* + \varepsilon/2$, and Corollary 10.12 with loss $\varepsilon/2$ gives the theorem. □

Remark 10.13. At the critical point the typical pattern has $\lambda(p) \approx X^{1/3}$, which shrinks the curvature box from S to about $S/X^{2/3}$. The weighted count of Lemma 10.6 shows that this average saving is available uniformly: the patterns with an unusually short vector are few enough to pay for their larger boxes.

11. The Lean formalization

The accompanying Lean 4 project uses toolchain `v4.35.0` and Mathlib `v4.35.0`. It proves every result of this paper, including the curve-counting input to Theorems 1.1 and 1.2, without additional hypotheses. It also proves the intermediate bounds listed in Section 1; of these, only the log-log refinement takes hypotheses, namely two published analytic estimates.

The strongest bound has the following statement:

```
theorem P_le_lattice_optimal :
    ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ,
    0 < b → b < a → (P a b : ℝ) ≤
    C * (a : ℝ) ^ ((471 : ℝ) / 1559 - 6 * Real.sqrt 105 / 7795 + ε)
```

The order of the quantifiers makes the uniformity explicit: C may depend on ε, but not on a or b. Here `P a b` is defined by well-founded recursion,

$$P(a,0) = 0, \qquad P(a,b) = 1 + P(a, a \bmod b) \quad (b > 0),$$

so it counts exactly the positive terms of the orbit. The formal orbit is indexed from zero, as in this paper.

The following table gives the public declarations; namespaces and the `.lean` extension are omitted. The repository README gives the fuller map from proof steps to modules.

Result	Declaration	Module
Theorem 1.1, γ_*	`P_le_lattice_optimal`	`LatticeJointMain`
Theorem 1.1, 153/520	`P_le_153_520`	`LatticeJointMain`
Theorem 1.2, α_*	`P_le_joint_optimal`	`JointLiftMain`
Theorem 1.2, 2201/7480	`P_le_2201_7480`	`JointLiftMain`
Theorem 1.2, 128/435	`P_le_128_435`	`JointLiftMain`
Theorem 1.3	`P_le`	`Main`
Intermediate, 23/77	`P_le_23_77`	`TwentyThreeMain`
Intermediate, 11/37	`P_le_11_37`	`ElevenMain`
Intermediate, 8/27	`P_le_8_27`	`EightMain`
Intermediate, 45/152	`P_le_45_152`	`FortyFiveMain`
Intermediate, log-log	`P_le_loglog`	`Hooley`
Theorem 1.4	`P_le_rises`	`RisesMain`
Theorem 7.2	`hankelInv_pos_int`	`Hankel`

The identifications of the first four invariants in Theorem 7.2 are proved in the same module as `hankelInv_one` through `hankelInv_four`.

For the curve theorem (23), the separate library `ccdn-dimension-growth` proves the real-B version, including the plane estimate, degree-preserving projection and box rescaling. The Pierce development imports it as a proved theorem and applies it to the defining prime ideals of its curves, using the same affine Hilbert degree as in Section 9. Thus the curve count is a proved dependency, not a hypothesis of the Pierce bound.

The proof of Theorem 1.2 is assembled in `PoweredJointBound`. The identities of Lemma 9.1(1) are proved in `FixedDividendIdentity` and `FixedDividendCurveEquations`, and the cover by at most twenty-four curves in `FixedDividendComponentCount`. The other main ingredients are the parity cover, the ordinary and conic gap counts, the graph-degree lower bounds, the powered pattern count and the window estimates; the repository README maps them to their modules. The transitive proof dependencies of the three public statements in `JointLiftMain` are audited to exclude the weighted-lattice counting modules of the 45/152 bound. The proof of Theorem 1.1 is assembled in `LatticeJointBound` and `LatticeJointOptimization`: Lemma 10.7 is `wcoord_spec` in `PivotExterior`, Lemma 10.9 is in `PivotCurvature` and `FixedDividendLatticeCoordinates`, Lemma 10.10 is in `LinearJointGraph`, and Lemma 10.6 is in `LatticeJointWeights`. A second audit checks that its two public statements reach, among the weighted counts, exactly Proposition 10.5 (`short_count`) and the reciprocal weight bound of Lemma 10.6. The intermediate bound `P_le_loglog` takes the propositions `NairTenenbaumDelta` and `BretecheTenenbaumMean` as hypotheses. They state the short-sum bound of [9] for the Erdős–Hooley function over intervals $(x, x+y]$ with $y \geq x^{1/4}$, and the mean-value bound of [2], including their existential constants and thresholds.

The file `Axioms.lean` audits the thirteen public statements and their proof dependencies. The only logical axioms reported are `propext`, `Classical.choice` and `Quot.sound`, the standard axioms used by Mathlib. The proof library contains no `sorry` and uses no `native_decide`. The separate Mathlib-only `Challenge.lean` states the thirteen results with deliberate theorem holes for comparison with the proved `Solution.lean`; neither module imports the other. Build instructions and the scope of the verification checks are recorded in the README.

Many local identities are checked by polynomial normalization over $\mathbb{Z}$, then transferred to the natural-number orbit after proving that the relevant differences are nonnegative. The counting arguments bound the number of parameter values and then the number of states with each value. Determinant positivity, integrality, curve reconstruction and the valuation

arguments are proved within the same development. The divisor estimate in Lemma 2.2 uses `DivisorPowerBound.lean` from `openai/math` [10] (Apache-2.0), with unchanged proof bodies and adapted imports and module visibility. The required Hadamard inequality is proved in `Hadamard.lean`.

12. Limits of the method

The threshold 5/17. In the proofs of Corollaries 9.14 and 10.12, three contributions give exactly the exponent 5/17, up to the arbitrarily small loss: the pattern count X^5 (from components that are not retained and from conics), the exceptional patterns, and windows in which the line-exclusion condition (18) fails. None of them involves the added coordinates W and J. So improving the coordinate embedding alone cannot go below $5/17 = 0.2941176\ldots$, which γ_* exceeds by about $1.1 \cdot 10^{-4}$. A stronger degree bound for the powered graphs may bring the exponent closer to 5/17; going below it requires a sharper count of the patterns that actually occur along an orbit. No claim of optimality for the Pierce iteration is made.

The earlier arguments. Proposition 5.3 removes the divisor loss from the local bound, but on its own the exponent is still 3/10. The local bound $T \ll a^{1/2}A^{-1/3}$ and the drop bound $T \ll A^{1/2}$ (up to the divisor factor) both allow $T \asymp a^{3/10}$ at $A \asymp a^{3/5}$. Even an ideal short-interval divisor estimate, removing the divisor factor entirely, leaves this balance unchanged. So sharpening constants or improving the divisor counts in that argument cannot give a logarithmic bound. To improve that argument, one needs to restrict *which* configurations of the parameters (u, v, w, z, D, E, F) can occur together along a single orbit. Theorem 1.4 is one restriction of this kind: a long orbit must have many rises in its quotient increments. Theorem 7.2 supplies the natural candidates for further ones, the higher invariants I_r. The patterns of Section 8 use I_4 in exactly this way.

The intermediate bounds count the states within each pattern using only the linear and quadratic relations among the increments. With patterns at four consecutive states one has thirteen integers and as many equations as unknown increments, so one might hope for $O(1)$ configurations per pattern. But the four-state equations have positive-dimensional families of solutions. For instance, the Newton coefficients $(\alpha, -3\alpha-1, 3\alpha+3, -\alpha-3, 1)$, with coefficient polynomial $(z-1)^3(z-\alpha)$, at every state, together with constant increments $k_j = t$, satisfy all the linear and quadratic relations for every t. Such families are excluded only by the remainder equations, which those counts discard. The fixed-dividend curve equations of Section 9 retain this information.

There is no obvious monotonicity to exploit. The curvatures D_i of successive blocks need not decrease. For $a = 13242$ the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132,\ \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$. The curvatures of the first five blocks are $24, 12, 6, 111, 30$, all for blocks that lie within a factor of two.

Appendix A. Conic components

In this appendix we prove Proposition 9.9. Let C be a nonexceptional component of degree two. A component that is not retained already contributes at most $404 \cdot 2 = 808$ starts, by Lemma 9.2. We may therefore assume that the component C is retained. If it contains at most two free-node triples of good starts, there is nothing to prove. In the remaining case we first construct a parameter whose value is an integer at every such triple.

Lemma A.1 (A bounded integral parameter)**.** *Let C be a retained nonexceptional degree-two component containing the free-node triples of at least three good starts. After permuting the*

three free-node coordinates to (x, y, z), there are an integer $\Delta_c \neq 0$ and polynomials $\xi_0, \xi_1, \xi_2 \in \mathbb{Z}[t]$ such that

$$\deg \xi_j \leq 2, \qquad |\Delta_c|,\ \max_{j,k} |[t^k]\xi_j| \leq (aX)^{O(1)}.$$

The map

$$t \longmapsto \big(\xi_0(t), \xi_1(t), \xi_2(t)\big)/\Delta_c$$

parametrizes C over $\mathbb{Q}$. Its inverse takes an integer value at every integer point of C, and is injective on those points.

Proof. Every free node of a good start has absolute value at most a. Three distinct such triples admit an integer plane

$$\lambda_z z + \lambda_x x + \lambda_y y + \lambda_1 = 0, \qquad |\lambda_z|, |\lambda_x|, |\lambda_y| \leq 8a^2, \qquad |\lambda_1| \leq 24a^3.$$

For example, take a cross product of two differences to obtain the normal; if that cross product vanishes, a nonzero perpendicular vector with the same bounds suffices. Put $K_{\mathrm{p}} = 24a^3$. A plane not containing an integral curve of degree two meets it in at most two distinct points. Thus this plane contains C. Choose the eliminated coordinate z with $\lambda_z \neq 0$.

The six rational pattern coordinates (9) have common denominator $N_0 N_1 \neq 0$ and respective integer numerators

$$N_1^2,\ u_1 N_1,\ u_2 N_1,\ N_2 N_0,\ v_1 N_0,\ v_2 N_0,$$

each of absolute value at most X^2. Expanding the quadratic f, its homogeneous part is a diagonal quadratic plus the square of a linear form. Its three diagonal numerators have denominator $N_0 N_1$, and its three linear numerators have denominator $(N_0 N_1)^2$; all six numerators have absolute value at most $12X^4$. Substitute $z = -(\lambda_x x + \lambda_y y + \lambda_1)/\lambda_z$ and multiply by $(N_0 N_1 \lambda_z)^2$. The resulting plane equation is

$$\alpha x^2 + \beta xy + \gamma_c y^2 + \ell_x x + \ell_y y + \ell_0 = 0, \tag{26}$$

with integer coefficients of absolute value at most $K = 10(12X^4)^2 K_{\mathrm{p}}^2$. Indeed, each coefficient is a sum of products of two quantities bounded by $12X^4$ and two plane coefficients bounded by K_{p}; expansion gives the stated factor 10. The restriction of f to the plane is nonzero: otherwise the plane would be a component of the irreducible pattern quadric. It has degree two because it contains C, and hence defines C in that plane.

The pullback of the hyperplane at infinity to the normalization of C has degree two. Corollary 9.4 says that each of its positive multiplicities is even. Counting residue degrees, there is therefore just one branch over infinity, of multiplicity two and residue degree one. Consequently the leading binary quadratic in (26) is a nonzero rational scalar times the square of a rational linear form. Indeed, the normalization map is finite and surjective, so the projective conic meets the line at infinity only at the image of this rational point. Its leading binary quadratic therefore has a single rational zero, of multiplicity two. Interchanging x and y if necessary gives

$$\alpha \neq 0, \qquad \beta^2 - 4\alpha\gamma_c = 0.$$

Set

$$\kappa_c = 2\beta\ell_x - 4\alpha\ell_y, \qquad \epsilon_c = \ell_x^2 - 4\alpha\ell_0, \qquad t = 2\alpha x + \beta y + \ell_x.$$

Completing the square in (26) gives

$$t^2 = \kappa_c y + \epsilon_c. \tag{27}$$

Here $\kappa_c \neq 0$. If it were zero, a rational point of C with parameter t_0 would give $\epsilon_c = t_0^2$, and the equation would factor as $(t - t_0)(t + t_0) = 0$. Integrality would then place C in an affine line, contradicting its degree.

We can therefore solve explicitly:

$$y = \frac{t^2 - \epsilon_c}{\kappa_c}, \qquad x = \frac{t - \beta y - \ell_x}{2\alpha}, \qquad z = -\frac{\lambda_x x + \lambda_y y + \lambda_1}{\lambda_z}.$$

Substitution verifies the conic and plane equations for every rational t, and the displayed linear expression in x, y recovers t. In particular, t is an integer whenever (x, y, z) is an integer point. Distinct points have distinct parameters. Taking $\Delta_c = 2\alpha\kappa_c\lambda_z$ clears all three denominators. The inequalities

$$|\kappa_c| \le 6K^2, \qquad |\epsilon_c| \le 5K^2, \qquad |\beta\epsilon_c - \ell_x\kappa_c| \le 11K^3$$

give $|\Delta_c| \le 12K_{\mathrm{p}}K^3$ and the coefficient bounds $33K_{\mathrm{p}}K^3$ for the three resulting quadratic numerators. For good starts one also has $|t| \le 4Ka$, although the counting argument below does not need a parameter interval. □

Lemma A.2 (Properness and coefficient height)**.** *In the parameter of Lemma A.1, write*

$$E_0(t) = \frac{F_E(t)}{G_E(t)}, \qquad F_E, G_E \in \mathbb{Z}[t], \qquad \gcd_{\mathbb{Q}[t]}(F_E, G_E) = 1.$$

The polynomials may be chosen so that

$$\deg G_E - \deg F_E = 3, \qquad \max(\deg F_E, \deg G_E) \le D_0, \qquad \max_k \big(|[t^k]F_E|, |[t^k]G_E|\big) \le C_0 a^{k_0},$$

for absolute constants D_0, C_0, k_0. In particular, G_E is nonconstant. At the parameter t of every good start, $G_E(t) \neq 0$ and $F_E(t)/G_E(t) = I_3(i)$ is an integer.

Proof. The explicit parametrization identifies C with the affine line and its smooth projective model with $\mathbb{P}^1$. Its point at infinity has hyperplane multiplicity two. Lemma 9.6 therefore gives

$$\mathrm{ord}_{t=\infty} E_0 = 3.$$

The function is nonzero, since it is positive at a good start. For a reduced rational function on $\mathbb{P}^1$, its order at infinity is $\deg G_E - \deg F_E$, proving properness.

We next bound the coefficients uniformly in the component, including after cancellation. Retain $K_{\mathrm{p}} = 24a^3$ and $K = 10(12X^4)^2K_{\mathrm{p}}^2$ from the preceding proof, and define

$$K_{\mathrm{h}} = 1 + a + X + 33K_{\mathrm{p}}K^3.$$

The six rational pattern coordinates, the three coordinate polynomials ξ_j/Δ_c, and the fixed dividend a have the common nonzero denominator $N_0N_1\Delta_c$. Their ten numerator polynomials have degree at most two and coefficient height at most K_{h}^4; the common denominator has absolute value at most K_{h}^4 as well. For instance, the coordinate numerators are $N_0N_1\xi_j$, while the numerator for a is $aN_0N_1\Delta_c$.

The cofactor formula defining E_0 is a quotient of two fixed polynomials in these ten inputs, with fixed rational coefficients. Clear those fixed coefficients once, then substitute the ten polynomials and clear a fixed power of their common denominator. This gives integer polynomials $\tilde{n}, \tilde{d}$, both nonzero, of bounded degree and coefficient height at most CK_{h}^r, where the degree bound, C and r are absolute. This assertion follows directly from polynomial multiplication: both the number of factors and the number of summands are fixed by the displayed cofactor formula.

Choose a primitive $\Pi \in \mathbb{Z}[t]$ representing $\gcd_{\mathbb{Q}[t]}(\tilde{n}, \tilde{d})$. By Gauss's lemma, Π divides both $\tilde{n}$ and $\tilde{d}$ in $\mathbb{Z}[t]$. Thus $F_E = \tilde{n}/\Pi$ and $G_E = \tilde{d}/\Pi$ are integer polynomials with the same ratio, coprime over $\mathbb{Q}$, and with degrees no larger than those of $\tilde{n}, \tilde{d}$. For a nonzero integer polynomial of degree at most D_0 and coefficient height $\mathcal{H}$, every integer polynomial factor has coefficient

height at most $2^{D_0}(D_0+1)\mathcal{H}$. To see this, the Mahler measure of an integer cofactor is at least one, the original polynomial's Mahler measure is at most $(D_0+1)\mathcal{H}$, and each coefficient of a factor is bounded by 2^{D_0} times its Mahler measure. Thus cancellation preserves a fixed polynomial bound in K_{h}.

Finally, $K_{\mathrm{p}} = 24a^3$ and $X \le 10^{16}a^7$ show that K_{h} is bounded by a fixed power of a. Absorbing the fixed exponents and constants gives the asserted $C_0a^{k_0}$ bound. The original reconstruction denominator is nonzero at every good start. Its cleared polynomial $\tilde{d}$ is therefore nonzero at the corresponding parameter, and so is its factor G_E. The reconstruction identity then gives $F_E(t)/G_E(t) = I_3(i) \in \mathbb{Z}$. □

Lemma A.3 (Integer values of a rational function)**.** *Let $F_E, G_E \in \mathbb{Z}[t]$ be coprime over $\mathbb{Q}$, of degrees at most D_0, with G_E nonconstant and coefficient heights at most $C_0a^{k_0}$. For every $\eta > 0$, the number of integers t such that $G_E(t) \neq 0$ and $F_E(t)/G_E(t) \in \mathbb{Z}$ is $O_\eta(a^\eta)$. The implied constant depends only on η, D_0, C_0, k_0.*

Proof. The integer resultant $R = \mathrm{Res}(F_E, G_E)$ is nonzero. The adjugate of the Sylvester matrix gives $U, V \in \mathbb{Z}[t]$ with $UF_E + VG_E = R$. At any parameter in question, $G_E(t) \mid F_E(t)$, and hence $G_E(t) \mid R$. There are $2\tau(|R|)$ possible signed values of $G_E(t)$. For each such value v, the nonzero polynomial $G_E - v$ has at most $\deg G_E \le D_0$ roots. Thus the number of parameters is at most

$$2D_0\tau(|R|).$$

The Sylvester matrix has size at most $2D_0$, and its entries are bounded by $C_0a^{k_0}$. Expanding its determinant gives

$$1 \le |R| \le (2D_0)!\big(C_0a^{k_0}\big)^{2D_0}.$$

The divisor estimate $\tau(n) \ll_\varepsilon n^\varepsilon$, with ε chosen in terms of η and the fixed degree and height exponents, proves the claim. □

Proof of Proposition 9.9. Apply the last lemma to the reconstructed invariant E_0. The parameter is injective on free-node triples, and reconstruction of the strictly increasing q_i makes those triples injective in the good starts. Each retained nonexceptional conic therefore contributes $O_\eta(a^\eta)$ starts. The components with at most two free-node triples of good starts and the nonretained components were already bounded absolutely. Apply the conic estimate with $\eta/2$ and the pattern estimate $O_{\eta/2}(a^{\eta/2}X^5)$. Since there are at most twenty-four components per pattern, all nonexceptional conics together contribute $O_\eta(a^\eta X^5)$. □

Appendix B. The upper bound for graph degrees

Proof of the upper bound in Lemma 9.12. Here only total polynomial degrees matter, not the heights of the pattern-dependent coefficients. Write the cofactor formula for the reconstructed invariant as $E_0 = E_{\mathrm{n}}/E_{\mathrm{d}}$, as in the proof of Lemma 9.2. Each entry of the moment matrix has degree at most four in the nodes. Expanding its 8×8 cofactors gives, in particular, the uniform bound $\deg c_j \le 40$, so $\deg E_{\mathrm{n}}, \deg E_{\mathrm{d}} \le 240$. Since H_3, H_3' have degree at most three and the gaps have degree at most one, the three curvature coordinates have a common denominator

$$\Omega = E_{\mathrm{d}}H_3k_1, \qquad \deg\Omega \le 244,$$

and numerators $\mathcal{N}_0, \mathcal{N}_1, \mathcal{N}_2$ of degree at most 244. The numerator for J with this denominator is $\mathcal{N}_3 = J\Omega$, of degree at most 247. The denominator is nonzero on C, since it is nonzero at a good start.

Put $\bar{m} = m + n$. Represent the powered coordinates with denominator $\Omega^{\bar{m}}$ and numerators

$$\mathcal{N}_j^m\Omega^{\bar{m}-m} \quad (0 \le j \le 2), \qquad \mathcal{N}_3^n\Omega^{\bar{m}-n}.$$

All these numerators have degree at most $247\bar{m}$. The original node coordinates, written with the same denominator, have numerators of degree at most $244\bar{m}+1$. It follows that multiplying a polynomial of degree at most t in the graph coordinates by $\Omega^{\bar{m}t}$ gives a polynomial in the original nodes of degree at most $(247\bar{m}+1)t$. This operation is injective on the corresponding spaces of functions, because Ω is nonzero in $\mathbb{Q}(C)$. Comparing their eventual affine Hilbert functions gives

$$\delta t + O(1) \le d(247\bar{m}+1)t + O(1),$$

and hence the asserted upper bound. In the unpowered case use Ω directly: the four new numerators have degree at most 247, and the original coordinates have numerators of degree at most 245. The same comparison gives $\delta \le 247d$. □

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Email address: `jason@jasondavies.com`